

Network UPS Tools Change Log

NUT project community contributors

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.8.5.792 2.8.5.818-818+g1e02b3869	2026-07-21	Current release snapshot of Network UPS Tools (NUT).	
2.8.5	2026-04-07	Some changes to docs and recipes, especially for parallel builds; introduced a way to build with private shared libraries to save space. Updated NUT SEMVER definition some more, added a normalization and comparison script. Some rounds of code-hardening project. Added reference RPM/DEB recipes to test NUT iterations on OBS. Numerous driver updates, some new ones introduced.	JK
2.8.4	2025-08-07	Fixed a few regressions introduced by release v2.8.3. Some changes to docs and recipes, especially for parallel builds. Updated NUT SEMVER definition some more. Some rounds of code-hardening project. Numerous driver updates, some new ones introduced.	JK
2.8.3	2025-04-07	Some changes to docs and recipes, libupsclient API and functionality. Updated NUT SEMVER definition and added scripting around it. Groundwork for vendor-defined status and INSTCMD buzzwords like "ECO". Fixed some regressions and added improvements for certain new device series.	JK
2.8.2	2024-04-01	Some changes to docs and recipes, libnutscan API and functionality. Added nutconf (library and tool). Fixed some regressions and added improvements for certain new device series.	JK
2.8.1	2023-10-31	Some changes to API, docs and recipes, in particular to simplify local builds and tests (e.g. to help end-users check if current NUT codebase trunk has already fixed an issue they see with a packaged installation). Revived NUT for Windows effort, further improved other OS integrations. NUT became reference for "UPS management protocol", Informational RFC 9271. Documentation files refactored to ease maintenance. More drivers and new driver categories introduced.	JK

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.8.0	2022-04-26	Change of maintainer. Many changes to API, docs (both style and content), and recipes, with a stress on non-regression test-ability, run-time debug-ability, general codebase maintainability, as well as OS integrations (notably nut-driver-enumerator for systemd and SMF service instance maintenance). Added a lot in area of CI support and documented pre-requisite package lists for numerous platforms, and CI agent set-up. Added libusb-1.x support and many new driver categories (and drivers), and daisychain device connection support. Instant commands enhanced with TRACKING to enable protocol-based waiting for completion of a particular INSTCMD or SET operation.	JK
2.7.4	2016-03-09	NUT variables namespace updated, in particular for outlet groups, alarms and thresholds, ATS devices, and battery.charger.status to supersede CHRG and DISCHRG flags published in ups.status readings. NUT network protocol extended with NUMBER type; some API changes.	AQ
2.7.3	2015-04-22	Documentation revised, including some API changes. Added NUT DDL links. NUT variables namespace updated.	AQ
2.7.2	2014-04-17	The nut-website project was offloaded into a separate repository. FreeDesktop HAL support was removed (obsoleted in GNOME consumer). Introduced nutdrv_atcl_usb driver.	AQ
2.7.1	2013-11-19	NUT source codebase migrated from SVN to Git (and from Debian infrastructure to GitHub source code hosting). jNut binding split into a separate project. Introduced libnutclient (C++ binding), al175, apcupsd-ups and nutdrv_qx drivers, Mozilla NSS support for simpler licensing than OpenSSL, and a newer apcsmart implementation. Documentation support enhanced with a spell checker, contents massively updated to reflect project changes.	CL

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.6.5	2012-08-08	New macosx-ups driver, new implementation of mge-shut driver. NUT variables namespace updated. Docs cleaned up and revised.	AQ
2.6.4	2012-05-31	New NUT network protocol commands (LIST CLIENTS, LIST RANGE and NETVER), and socket protocol commands (ADDRANGE, DELRANGE). NUT variables namespace updated. Introduced nut-recorder tool.	AQ
2.6.3	2012-01-04	No substantial changes to documentation.	AQ
2.6.2	2011-09-15	Introduced nut-scanner tool and nut-ipmipsu driver, systemd support, and a new apcsmart implementation.	AQ
2.6.1	2011-06-01	Introduced default.* and override.* optional settings in ups.conf, an ups.efficiency report, and outlet.0 special handling.	AQ
2.6.0	2011-01-14	First release of AsciiDoc documentation for Network UPS Tools (NUT).	AQ

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1 Introduction

The primary goal of the Network UPS Tools (NUT) project is to provide support for Power Devices, such as Uninterruptible Power Supplies, Power Distribution Units and Solar Controllers.

2 Very detailed Change Log

This document intends to detail the change log for relatively recent work (roughly since the source code was tracked in Git).

Note

This change log section represents git commits in range `v2.8.5..HEAD` (commits `0e051f9f6..1e02b3869`).

2.1 2026-07-20 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: update for refactor of `background()` method [#3437]
- drivers/main.c: fix comment style to old C [#3437]
- NEWS.adoc, docs/nut.dict, drivers/nutdrv_qx_voltronic-qs.c: drivers/nutdrv_qx_voltronic-qs.c, NEWS.adoc: bump VOLTRONIC_QS and reword the news entry [#3521]
- drivers/nutdrv_qx_voltronic-qs-hex.c, NEWS.adoc: fix format string for `ups.load` [#3532]
- INSTALL.nut.adoc: update with a larger note about packages lagging behind development tip [#3531] Also remove references to package maintainers being part of core team, there was a bit of turbulence in the past years, such notes get obsolete. . .

2.2 2026-07-19 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/app/NUT-Monitor-py2gtk2.in, scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in, scripts/python/app/NUT-Monitor-*.in: refactor old `handler teardown` with new `graceful` and `harsher disconnect()` handling [#3509]
- scripts/python/app/NUT-Monitor-py2gtk2.in, scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in, scripts/python/module/PyNUT.py.in: scripts/python/app/NUT-Monitor-*.in, scripts/python/module/PyNUT.py.in: refactor with a `graceful` and `harsher disconnect()` handling [#3509]
- scripts/python/app/NUT-Monitor-py2gtk2.in: handle lack of `pynotify` more gracefully [#3509]
- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: scripts/python/app/NUT-Monitor-py3qt{5,6}.in: refactor `gui_status_message()` with `try/except` handling [#3509]
- scripts/python/app/NUT-Monitor-py2gtk2.in: `gui_status_message()`: fix indentation [#3509]
- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: scripts/python/app/NUT-Monitor-py3qt{5,6}.in: fix "Broken pipe" and possible stall in reconnection attempts [#3509]
- scripts/python/module/PyNUT.py.in: refactor with an error-logging `__send()` [#3509]
- scripts/python/module/PyNUT.py.in: `__read_until()`: handle the case of un-received chunk [#3509]
- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: scripts/python/app/NUT-Monitor-py3qt{5,6}.in: mark the templates as nominally executable

2.3 2026-07-17 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/mge-hid.c, NEWS.adoc: bump DRIVER_VERSION and update news about CHRG status suppression [#3518, #3519]
-

2.4 2026-07-15 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc, scripts/python/app/NUT-Monitor-py2gtk2.in: scripts/python/app/NUT-Monitor-py2gtk2.in: propose a fix for re-connection modeled after [PR #3523](#) for Python3 siblings [[#3509](#)] Co-Authored-By: Programer <developer.sorli@gmail.com>
- NEWS.adoc: NUT-Monitor updated with automatic reconnection [[#3509](#), [#3523](#)]
- NEWS.adoc, drivers/blazer_usb.c, drivers/nutdrv_atcl_usb.c, drivers/nutdrv_qx.c, drivers/powercom-hid.c, drivers/richcomm_usb.c, drivers/riello_usb.c, drivers/tripplite_usb.c: drivers/*.c, NEWS.adoc: play safer with temporary buffers, pre-zero some [[#3529](#)]
- clients/upsmon.c: cast assignments into temporary argv[] [[#3499](#)]
- tests/NIT/upssched.conf.in: fix path to upssched-cmd sample [[#3499](#), [#1711](#)]
- tests/NIT/nit.sh: we no longer support NOTIFYCMD with shell snippets in one token [[#3499](#)]
- clients/upsmon.c: upsmon_cleanup(): NULLify freed allocations; free cert-related data [[#3499](#)]
- NEWS.adoc, clients/upsmon.c, clients/upssched.c: clients/upssched.c: add an atexit(clean_exit) handler similar to that in upsmon [[#3499](#)]
- NEWS.adoc, UPGRADING.adoc, clients/upsmon.c, clients/upssched.c, conf/upsmon.conf.sample.in, conf/upssched.conf.sample.in, docs/man/upsmon.conf.txt, docs/man/upssched.conf.txt: upsmon, upssched: extend SHUTDOWNCMD, NOTIFYMSG and CMDSCRIPT to handle command-line args as separately quoted tokens in the directive [[#3499](#)]

2.5 2026-07-14 Jim Klimov <jimklimov+nut@gmail.com>

- common/wincompat.c, drivers/dstate.c, clients/upssched.c: CreateNamedPipe(): add PIPE_REJECT_REMOTE_CLIENTS flag to only work with local host clients [[#3527](#)]
- common/wincompat.c: syslog() shim: protect against re-entry if we upslog()/upsdebug() from this method [[#3527](#)]
- common/wincompat.c: syslog() shim: do not flood stderr with repetitive SKIP: can't open existing event log NAMED_PIPE messages [[#3527](#)]
- clients/upssched.c: help(): fix reference to man page
- clients/upssched.c, NEWS.adoc: fix notify_msg from CLI when other options are present [[#3525](#), [#3105](#)]
- tests/NIT/nit.sh: do not die() in certificate prerequisite checks if we can follow up, logging the error in a way that seems too fatal to warnings parsers [[#1711](#)]
- ci_build.sh, tests/NIT/nit.sh: simplify cut_filter() [[#3526](#), [#3109](#)]
- Do not add optional dependency on cut * Consider starting AND ending characters of input hash string for more randomness
- NEWS.adoc, ci_build.sh, tests/NIT/nit.sh: ci_build.sh, tests/NIT/nit.sh: use shorter hash strings for cache dirs [[#3526](#), [#3109](#)]

2.6 2026-07-13 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upssched.c, common/wincompat.c, NEWS.adoc: use WIN32 security attributes for listening pipes [[#3525](#)] Replicate what was done for dstate.c in [PR #3479](#). Co-authored-by: mbv06 <mbv06.dev@gmail.com>
- common/wincompat.c, include/wincompat.h, drivers/dstate.c: move init_pipe_security() to be reusable [[#3479](#), [#3525](#)] Co-authored-by: mbv06 <mbv06.dev@gmail.com>
- include/wincompat.h: move include guard to be after heading comments
- configure.ac: if CPPUNIT seems present but won't build, suggest if C++ version could be the problem
- scripts/perl/UPS/Nut.pm.in, scripts/perl/test_nutclient.pl, scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py: fix parsing of CERTIDENT NAME/PASS from nutauth.conf files [[#3329](#)]

- clients/authconf.c, clients/nutclient.cpp, scripts/perl/UPS/Nut.pm.in, scripts/python/module/PyNUT.py.in: consistently log seen unsupported keywords in nutauth.conf files [#3329] ... this can well be a typo in parser code, help catch those.
- tests/NIT/nit.sh: revise generation of PEM files with NSS binary builds [#1733, #3329] Now we need the PEM files always (when we can make them) for PERL and Python test suites. Generation of JKS (e.g. for jNut) remains conditional - if we would maintain the test certificate cache.
- tests/NIT/nit.sh: fall back in nutauth-openssl.conf to use rootca.pem if no CERTHASH.0 links were populated [#1711, #3329]
- tests/NIT/nit.sh: actively not-enable "CERTFILE" in nutauth.conf if upsmon.pem is missing [#1711, #3329]
- tests/NIT/nit.sh: reshuffle determination of DO_USE_NIT_TESTCERT_CACHE and log different outcomes [#1711, #3329]
- tests/NIT/nit.sh: at least report that upsmon.pem is missing and tests may fail, if we did not get one fallbacked for NSS builds [#1711]
- tests/NIT/nit.sh: annotate failure messages for easier search in make-oriented logs [#1711]
- ci_build.sh: annotate default-all-errors failure messages for easier search in make-oriented logs [#1711]

2.7 2026-07-12 Jim Klimov <jimklimov+nut@gmail.com>

- configure.ac: check for systemctl earlier, so it feeds into optional warning about lack of systemdsystemunitdir

2.8 2026-07-09 Jim Klimov <jimklimov+nut@gmail.com>

- clients/Makefile.am, clients/upsclient.c, clients/upsclient.h: clients/upsclient.{c,h}: keep UPSCONN_t size same for different SSL backends [#3331]

2.9 2026-07-05 Programer <developer.sorli@gmail.com>

- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: NUT-Monitor: Add automatic reconnection mechanism for Qt5 and Qt6 - Add automatic reconnection when connection to NUT server is lost - Implement proper teardown of old PyNUTClient instances - Use specific exceptions (socket.error, ConnectionError, EOFError, OSError, PyNUT.PyNUTError) - Prevent multiple simultaneous reconnection attempts - Add *was_online* flag to track connection state changes
 - Tested and confirmed working on both Qt5 and Qt6 Closes: #3509

2.10 2026-07-03 Stephen Horvath <s.horvath@outlook.com.au>

- docs/nut.dict: Add voltronic-qs *QI* command
- NEWS.adoc, data/driver.list.in: NEWS.adoc, driver.list.in: Update for voltronic-qs & PSDR800
- drivers/nutdrv_qx_voltronic-qs.c: nutdrv_qx: Voltronic-QS: Add QI command This command adds battery charge, runtime, input frequency, output current, and input transfer low/high values to the Voltronic-QS driver. Tested on a PowerShield PSDR800.

2.11 2026-07-01 strasharo <3337997+strasharo@users.noreply.github.com>

- drivers/mge-hid.c: mge-hid: suppress CHRGE on CC-mode devices when battery is fully charged UPS.PowerSummary.PresentStatus.C stays asserted in float/ maintenance mode on constant-charge (CC-mode) devices such as the Eaton 5E series. These devices do not expose the ABM HID path (UPS.BatterySystem.Charger.ABMEnable), so advanced_battery_monitoring stays ABM_UNKNOWN. eaton_abm_check_chrg_fun() then maps any Charging=1 report directly to CHRGE with no distinction between active charging and float mode. The result is ups.status stuck at OL CHRGE indefinitely, even when battery.charge is 100. Fix: when ABM is UNKNOWN or DISABLED and the UPS asserts Charging=1, check battery.charge before publishing CHRGE. If the battery is at 100% the charger is in float/maintenance mode, not actively charging, so return !chrg instead. Tested on Eaton 5E 650i (vendor 0463, product ffff, subdriver MGE HID 1.58, firmware 01.04.0016) where ups.status now correctly shows OL at full charge instead of OL CHRGE.

2.12 2026-06-30 Jim Klimov <jimklimov@gmail.com>

- tests/NIT/nit.sh: setenv_ssl_perl(): in case of darwin OS, neuter CERTVERIFY also in nutauth.conf [#1711]

2.13 2026-06-30 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c, clients/authconf.h: clients/authconf.h: do not require nut_stdint.h in public API [#3329]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: fix AuthConf::authconf_list to be actually a list, not vector [#3329] Memory-safer to extend and manipulate.
- clients/nutclient.cpp: SSLConfig::addCertHost(): avoid duplicates [#3329]
- tests/NIT/nit.sh: setenv_ssl_perl(): fix up paths on Windows in a new nutauth-openssl-perl-win.conf [#3329, #1711]
- tools/gitlog2changelog.py.in: fix unicode TypeError with python 2.7
- scripts/python/module/PyNUT.py.in: AI added "import getpass" needlessly [#3329]
- scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py.in: scripts/python/module/PyNUT.py.in: do not strip brackets from section names after creation (fix IPv6 address handling) [#3229, #1711]
- scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py.in: avoid Python 3 formatting strings [#3229, #1711] There is no reason for module to not work for older OS installations.
- tests/NIT/nit.sh: setenv_ssl_*(?): pass NUT_SSL=true for SSL builds even if skipping other settings due to nutauth.conf [#3329, #1711] May neuter later in the method (e.g. perl lacking SSL module).
- NEWS.adoc, clients/upsmon.c, clients/upsmon.h, common/nutconf.cpp, common/nutwriter.cpp, include/nutconf.hpp: clients/upsmon, nutconf.cpp, nutwriter.cpp, NEWS.adoc: introduce NOTIFY_SUSPEND_TIMEJUMP_UNEXPECTED [#3405]

2.14 2026-06-29 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/test_nutclient.py.in, scripts/perl/test_nutclient.pl, tests/cpptest-client.cpp: only set "NUT_SSL=True" from AuthConf NUT_FORCESSL if no value is passed as envvar [#3329, #1711]
- tests/cpptest-client.cpp: setUp(): fix env_NUT_CERTHOST_ADDR and env_NUT_CERTHOST_NAME [#3329, #1711]
- tests/cpptest-client.cpp: setUp(): fix env_NUT_CERTSTORE_PATH vs env_NUT_CAPATH (nss vs openssl backend) [#3329, #1711]
- tests/cpptest-client.cpp: setUpClientSSL(): trace incoming envvars and configs being applied [#3329, #1711]
- tests/cpptest-client.cpp: setUp(): fix env_NUT_CERTIDENT_NAME vs env_NUT_KEYFILE [#3329, #1711]
- tests/cpptest-client.cpp: revise certfile vs certpath handling [#3329, #1711]
- scripts/python/module/test_nutclient.py.in: no longer need detailed AuthConf debug in tests [#3329]
- scripts/python/module/PyNUT.py.in: fix readAuthConfFile() to strip spaces from KEY [#3329]
- scripts/python/module/PyNUT.py.in: fix readAuthConfFile() to create new section with stripped brackets [#3329]
- scripts/python/module/PyNUT.py.in: fix readAuthConfFile() to use open() not os.open() [#3329]
- scripts/python/module/PyNUT.py.in: revise logging in AuthConf getter, reader, merger [#3329]
- scripts/python/module/PyNUT.py.in: add PyNUT.AuthConf.str() method [#3329, #1711]
- scripts/perl/test_nutclient.pl: revise certfile vs certpath handling
- tests/NIT/nit.sh: refine NUT_SSL=false vs NUT_IGNORE_AUTHCONF [#1711, #3329]
- tests/NIT/Makefile.am: check-NIT-devel: we now depend on more clients [#1711]

2.15 2026-06-28 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: track AuthConf section name value without the brackets [#1711, #3329]
- scripts/perl/test_nutclient.pl: tone down AuthConf parser debugging now [#1711, #3329]
- tests/NIT/nit.sh: populate upsmon.pem for OpenSSL-only clients when generating NSS certs [#1711, #3311]
- scripts/perl/UPS/Nut.pm.in: fix key casing in readAuthConfFile() [#3329]
- scripts/perl/UPS/Nut.pm.in: more logging and comments to readAuthConfFile() [#3329]
- tests/NIT/nit.sh: generatecfg_nutauth(): produce a nutauth-openssl.conf for Perl and Python tests to work even with NSS C builds [#3329, #1711]
- scripts/perl/UPS/Nut.pm.in: rectify merge() logging [#3329]
- tests/NIT/nit.sh: generate nutauth.conf with both "localhost" and "127.0.0.1" entries for different queries [#3329, #1711]
- scripts/perl/UPS/Nut.pm.in, scripts/perl/test_nutclient.pl, scripts/python/module/test_nutclient.py.in: scripts/perl/UPS/Nut.pm.in, scripts/perl/test_nutclient.pl: rectify findAuthConf()/getAuthConf() logic and logging [#3329]
- docs/Makefile.am: ChangeLog.adoc-parsed: convert markdown links to asciidoc (grok dependabot commit comments) [dependabot/core#15228]
- NEWS.adoc: move ragtech entry to post-2.8.5 release

2.16 2026-06-26 juslex <66561713+juslex@users.noreply.github.com>

- scripts/upsdrvsvccctl/nut-driver-enumerator.sh.in, tests/nut-driver-enumerator-test—ups.conf, tests/nut-driver-enumerator-test.sh: nut-driver-enumerator: classify ragtech as serial media The ragtech driver talks to a USB CDC-ACM device node (/dev/tty-ACM*) through the serial layer. Classifying it as "serial" media lets the auto-generated nut-driver@<ups> service depend on systemd-udev / nut-udev-settle, so it waits for the device node to be published by udev before starting instead of racing the enumeration at boot. Requested by @jimklimov in [issue #3500](#). Add a [ragtech-serial] section to the enumerator self-test fixture and the matching entries in the affected expected outputs (device-name listing, config dump, and MEDIA/checksum listing). Full self-test passes (104 cases).

2.17 2026-06-26 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/test_nutclient.py.in: fix interpretation of ac.certfile [#3329]
- tests/cpptest-client.cpp: debug progress through authconf parsing [#3329, #1711]
- scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py.in: scripts/python/module/*: debug progress through authconf parsing
- scripts/perl/UPS/Nut.pm.in, scripts/perl/test_nutclient.pl: scripts/perl/*: debug progress through authconf parsing
- scripts/python/module/PyNUT.py.in: AuthConf: fix upper-/any- case expectations [#3329]
- scripts/perl/UPS/Nut.pm.in: readAuthConfFile(): fix upper-/any- case expectations [#3329]
- tests/NIT/nit.sh: report current SSLBACKEND in nutauth.conf [#1711]
- clients/nutclient.cpp: handle_authconf_args(): AuthConf::merge(): pepper with debug [#3329]
- clients/nutclient.cpp: handle_authconf_args(): set user name early (if known from section), so we can ignore it in set_authconf_val() [#3329]
- clients/nutclient.cpp: set_authconf_val(): fix upper-/any- case expectations [#3329]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{h,cpp}: introduce methods to dump AuthConf data for debug [#3329]

- clients/authconf.c: update comment [#3329]
- scripts/python/module/test_nutclient.py.in, scripts/python/module/PyNUT.py.in: reshuffle debugging to see into AuthConf handling [#3329, #1711]
- scripts/python/module/PyNUT.py.in: debug-log more details about SSL setup if it is enabled [#3329, #1711]
- clients/authconf.h, clients/nutclient.h, scripts/perl/UPS/Nut.pm.in, scripts/python/module/PyNUT.py.in: C, C++, Perl, Python AuthConf implementations: update comment about "certfile" field [#3329]
- conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: fix CERTFILE description ("client" not "server" here) [#3329]
- scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py.in: scripts/python/module/PyNUT.py.in: rename AuthConf "pass_" field to "password" [#3329]
- scripts/perl/UPS/Nut.pm.in, scripts/python/module/PyNUT.py.in: document meanings of AuthConf fields [#3329]
- scripts/python/module/PyNUT.py.in: fix back mis-merged edits [#3329]
- scripts/perl/test_nutclient.pl: reshuffle debugging to see into AuthConf handling [#3329, #1711]
- scripts/perl/UPS/Nut.pm.in: UPS::Nut→new(): support args hash with an AUTHCONF key to parse it and merge with other args [#3329]
- scripts/perl/UPS/Nut.pm.in: revise debug logging of AuthConf progress [#3329]
- clients/nutclient.cpp: parse_authconf_file(): upsdebugx() which file we are looking at [#3329]
- scripts/perl/test_nutclient.pl, scripts/python/module/PyNUT.py.in, scripts/python/module/test_nutclient.py.in, tests/cpptest-client.cpp: Fix C++, Perl, Python test runners to consider nutauth.conf [#3329] WIP
- scripts/perl/UPS/Nut.pm.in: refactor error/debug messaging from readAuthConfFile() [#3329]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{h,cpp}: add AuthConf operator= [#3329]

2.18 2026-06-25 Thanh Nguyen <imscopilot@outlook.com>

- data/driver.list.in: Add ARES AR630 to HCL

2.19 2026-06-24 dependabot[bot] <49699333+dependabot[bot]@users.noreply.github.com>

- .github/workflows/01-make-dist-worker.yml, .github/workflows/05-codeql.yml: build(deps): bump actions/cache from 5 to 6 Bumps [actions/cache](#) from 5 to 6. - [Release notes](#) - [Changelog](#)
 - [Commits](#) --- updated-dependencies: - dependency-name: actions/cache dependency-version: 6 dependency-type: direct:production update-type: version-update:semver-major ...
- .github/workflows/01-make-dist-worker.yml, .github/workflows/05-codeql.yml: build(deps): bump actions/checkout from 6 to 7 Bumps [actions/checkout](#) from 6 to 7. - [Release notes](#) - [Changelog](#)
 - [Commits](#) --- updated-dependencies: - dependency-name: actions/checkout dependency-version: 7 dependency-type: direct:production update-type: version-update:semver-major ...

2.20 2026-06-24 Jim Klimov <jimklimov+nut@gmail.com>

- tests/cpptest-client.cpp, scripts/perl/test_nutclient.pl, scripts/python/module/test_nutclient.py.in: use nututh.conf files if possible [#3329, #1711]
- tests/NIT/nit.sh: setenv_ssl_*(): skip work by default if nutauth.conf exists [#1711, #3329]
- scripts/perl/UPS/Nut.pm.in, scripts/python/module/PyNUT.py.in: revise splitting of host:port strings where host looks like numeric IPv6 [#3503]
- clients/nutclient.cpp: AuthConf::merge(): gracefully handle section name that starts with "@" [#3329]
- clients/nutclient.cpp, scripts/perl/UPS/Nut.pm.in, scripts/python/module/PyNUT.py.in: revise normalization of host strings that look like numeric IPv6 [#3503]
- clients/nutclient.cpp: update splitaddr() against libupsclient variant
- scripts/python/module/test_nutclient.py.in, tests/NIT/Makefile.am: scripts/python/module/test_nutclient.py: add testsuite_splitaddr() [#3329, #1711]
- scripts/python/module/PyNUT.py.in: allow to construct PyNUTClient without connecting right away; remember requested TRACKING option to apply whenever we do connect() [#3329, #1711]
- configure.ac, scripts/Makefile.am, scripts/perl/.gitignore, scripts/perl/UPS/{Nut.pm ⇒ Nut.pm.in}, tests/NIT/Makefile.am, tests/NIT/nit.sh: Convert UPS/Nut.pm into a .in template to handle @CONFPATH@ correctly [#1711, #3329]
- clients/authconf.c: parse bracketed IPv6 addresses correctly [#3329, #3503]
- tests/test_authconf.c: add tests to parse IPv6 addresses (brackets and colons!) [#3329, #3503]
- tests/test_authconf.c: test that (comment) text after a section name is ignored [#3329]
- tests/test_authconf.c: auto-account expected_sections (to check parsed num_sections) [#3329]
- tests/NIT/nit.sh: generatecfg_nutauth(): only handle case WITH_SSL_CLIENT_CERTIDENT ... if WITH_SSL_CLIENT != none [#1711]
- tests/NIT/nit.sh: move check for usability of certutil or openssl into prepare_NIT_certs() where we would inevitably use them [#1711]
- tests/NIT/nit.sh: fix indentation [#1711]

2.21 2026-06-23 juslex <66561713+juslex@users.noreply.github.com>

- NEWS.adoc, docs/man/ragtech.txt, docs/nut.dict, drivers/ragtech.c: ragtech: fix CDC-ACM startup (raw mode \+ initial-poll retry), battery scaling, va override Startup robustness (issue #3500, Easy Pro 3200 VA GT on a Raspberry Pi) -----
----- The reporter's unit failed every boot with "no reply to initial status poll", even though it speaks the family-10 protocol correctly (its 30-byte main range decodes cleanly against the OEM Supervise 8 readings). The root cause, pinned down from an strace \+ the probe script, was the serial line discipline, not the protocol:
- A freshly enumerated CDC-ACM tty comes up in canonical mode (ICANON). In that mode the kernel buffers incoming bytes until it sees a newline. The firmware's replies are raw binary and never contain one, so the bytes were held in the kernel and the first poll always timed out. The driver had been relying on the port already being raw — which happened to be true on the author's host only because its systemd unit ran "stty ... raw" in an ExecStartPre, masking the bug. Without that external step (e.g. a plain "upsdvctl start"), the driver never worked. upsdv_initups() now sets raw 8N1 mode itself: tcsetattr/clear ICANON et al./tcsetattr, leaving the baud untouched (CDC-ACM ignores it at the wire), with CLOCAL set and HUPCL cleared so neither the reconfigure nor the eventual close toggles DTR, and DTR/RTS are forced low afterwards. The previous "NEVER call tcsetattr" comment was unfounded: NUT's own ser_set_speed() configures raw mode via tcsetattr and is used by ~50 serial drivers, the reporter observed no DTR pulse / shutdown, and this driver's shutdown path is a register write, not a DTR edge. * upsdv_initinfo() now retries the wake-up handshake and first poll a few times before calling fatalx(). Right after a cold boot the device may have only just enumerated and the firmware can need a moment to answer. Retrying lets the driver come up cleanly under the stock NUT systemd units (which start it as soon as the device node appears) instead of depending

on a fatal exit plus a Restart=on-failure cycle to recover, which is noisy and only retries on a 5 s cadence. Combined with the raw mode above, no external stty, sleep, or .device binding is needed. * The firmware answers the handshake with a byte or two (e.g. 0xCA). Some models are half-duplex and drop a command sent while that reply is still in flight; the handshake reply is now drained before the first poll (which only works because raw mode actually delivers those bytes). Other fixes ----- * Compute battery.charge as raw / 2.55 so a full battery reports exactly 100 % instead of 100.2 % (drops the previous clamp). * Add a "va" ups.conf option to override the apparent-power rating. The model id byte is shared between product lines (a 3200 VA GT reports the same id as an Easy 2200 TI), so the driver cannot always pick the right VA from its table; the override corrects ups.power.nominal, ups.realpower.nominal and the computed ups.load. Verified: family-10 Easy 2000 TI on hardware (raw mode, handshake drain, battery 100.0, va=3200 override) and the 3200 VA GT by the reporter (full telemetry, matching Supervise 8). Driver bumped to 0.10, still DRV_EXPERIMENTAL.

2.22 2026-06-23 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: upscli_free_host_port_cert_data(): only defined for SSL-capable builds [#3329]
- clients/upsclient.c: upscli_free_host_port_cert_data(): consider const-ness of freed strings [#3329]
- clients/upsclient.c: upscli_free_host_port_cert(): consider builds without SSL [#3329]
- drivers/dummy-ups.c: refactor with upscli_authconf_update_conn_flags() [#3329]
- clients/upsc.c, clients/upsclient.c, clients/upsclient.h, clients/upscmd.c, clients/upsimage.c, clients/upslog.c, clients/upsrw.c, clients/upsset.c, clients/upsstats.c, tools/nut-scanner/scan_nut.c: tools/nut-scanner/scan_nut.c, clients/*.c: refactor with upscli_authconf_update_conn_flags() [#3329]
- clients/upsc.c, clients/upscmd.c, clients/upsimage.c, clients/upslog.c, clients/upsrw.c, clients/upsset.c, clients/upsstats.c: clients/*.c: only consider authconf default certverify and flags_ssl if positive [#3329]
- tests/NIT/nit.sh: on MSYS2, copy (not symlink) root CA cert hash-named files right away [#1711]
- tests/NIT/nit.sh: when transplanting prepared certs to/from cache, fall back to dereferencing symlinks if cp -prf failed [#1711]
- tests/NIT/nit.sh: TESTCERT_PATH_SEP: there was too much of a good thing for WIN32 runs [#1711]
- tests/NIT/nit.sh: use TESTCERT_PATH_SEP more diligently (e.g. on WIN32 runs) [#3331, #1711]
- tests/NIT/nit.sh: die() with a timestamp too

2.23 2026-06-23 Boris Shor <boris@bshor.com>

- NEWS.adoc, data/driver.list.in, docs/nut.dict, drivers/idowell-hid.c: idowell-hid: claim GoldenMate LiFePO4 (06da:ffff) without hijacking other Phoenixtec devices GoldenMate 1000VA/800W LiFePO4 packs reuse the shared Phoenixtec vendor ID 0x06da:0xffff with the same -BMS- firmware and HID descriptor as the existing iDowell 0x075d:0x0300 device, so they fall back to the generic liebert-hid handler (bogus battery.runtime, etc). The 0x06da VID is shared: mge-hid claims it for AEG PROTECT NAS and liebert-hid is its default sink. Probe order is mge → idowell → liebert, so idowell can intercept GoldenMate before liebert provided it does not grab the other 0x06da devices. - Add a named PHOENIXTEC_VENDORID define and use it for the 0x06da:0xffff device-table entry (feeds the udev/nut-scanner parser). - Gate idowell_claim() on the device's -BMS-/Smart-Battery strings, like mge_claim() does for AEG, returning 0 for non-GoldenMate 0x06da so they fall through to liebert-hid / mge-hid. - Bump IDOWELL_HID_VERSION to 0.21 and add a NEWS.adoc entry. - List both GoldenMate variants in data/driver.list.in. Fixes: <https://github.com/networkupstools/nut/issues/3501>

2.24 2026-06-22 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc, UPGRADING.adoc, clients/upsmon.c, clients/upssched.c, conf/upsmon.conf.sample.in, conf/upssched.conf.sample.in, docs/man/upsmon.conf.txt, docs/man/upsmon.txt, docs/man/upssched.conf.txt, docs/man/upssched.txt, docs/nut.dict: clients/upsmon.c, clients/upssched.c, docs: handle NOTIFYCMD and CMDSCRIPT values strictly as paths [#3499]

- docs/configure.txt: properly pass RPATH for linker not compiler in LDFLAGS [#3495]
- NEWS.adoc: update about recipe and code portability PRs [#3489, #3495, #3498]
- ci_build.sh: properly pass RPATH for linker not compiler in LDFLAGS [#3495]
- NEWS.adoc, clients/Makefile.am, clients/upsimage.c, clients/upsimagearg.h: clients/upsimage.c, clients/upsimagearg.h: adjust scaling for small (single-digit) ranges [#3469] Avoid lack of output when the range of a variable is small (e.g. 0-10) and division by zero occurred instead.
- NEWS.adoc, clients/upsc.c, clients/upscmd.c, clients/upsimage.c, clients/upslog.c, clients/upsmon.c, clients/upsrw.c, clients/upssched.c, clients/upsset.c, clients/upsstats.c: clients/*.c, NEWS.adoc: fflush() stderr and stdout before clean-up and exit, just in case [#3454]
- clients/upsclient.c: wherever we fail upscli_init*(), call upscli_cleanup() right away [#3454]
- clients/upsclient.c: pre-initialize SSL_CTX as NULL to be sure [#3454]

2.25 2026-06-21 Jim Klimov <jimklimov+nut@gmail.com>

- tests/.gitignore: GitIgnore tests/test_authconf if built [#3329]
- NEWS.adoc, docs/man/dummy-ups.txt, drivers/dummy-ups.c: dummy-ups: add NUT authconf support [#3329]

2.26 2026-06-21 Huang Rui <vowstar@gmail.com>

- NEWS.adoc, drivers/xppc-mib.c: feat(snmp-ups): map more xppc readings for Phoenixtec UPS cards - add battery.runtime, battery.voltage and input.frequency readings - add output.voltage.nominal and ups.firmware.aux readings - bump XPPC_MIB_VERSION to 0.41

2.27 2026-06-21 Doug Nazar <nazard@nazar.ca>

- server/upsd.c: AIX 64 uses a long for fd in pollfd
- ci_build.sh: Tell compiler to pass -R option to linker instead of a remark

2.28 2026-06-20 penghong123 <2835950872@qq.com>

- data/driver.list.in: Add some new UPS models

2.29 2026-06-20 Doug Nazar <nazard@nazar.ca>

- include/timehead.h: Provide function declaration when needed
 - drivers/powervar_cx_ser.c: AIX doesn't provide defines for those speeds.
 - clients/upsc.c, clients/upscmd.c, clients/upslog.c, clients/upsrw.c: Ignore unused parameters
 - Makefile.am: Switch to using -t to specify date as -d not always available
 - configure.ac: Fix %zu format check
 - configure.ac: Use lsgroup on AIX to check for valid groups
 - m4/nut_compiler_family.m4: Check enable version of warnings to see if valid GCC will silently accept warning disable options, but then if there is an issue, warn about the option. "cc1: note: unrecognized command-line option *-Wno-unknown-warning-option* may have been intended to silence earlier diagnostics"
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- scripts/python/module/Makefile.am: Don't try to copy permissions When the source files are on a different filesystem it's possible for the copy to fail. Seen on FreeBSD, when trying to copy from an NFS mount it would fail trying to copy the ACLs.
- tools/Makefile.am: Use full path for generated files When using relative paths Make will use VPATH to find the file in the source directory. If out of date, it will then try to update that file instead and fail on a read-only source directory.
- include/common.h, tools/nut-scanner/scan_eaton_serial.c: Add check for older IBM compilers. Newer compilers are based on Clang and errors on GCC.

2.30 2026-06-19 Jim Klimov <jimklimov+nut@gmail.com>

- tools/nut-scanner/scan_nut.c: follow up on NUT authconf support when scanning "old nut" servers [#3329]
- clients/upsclient.c, clients/upsclient.h: clients/upsclient.{c,h}: add pthreads support for first_host_cert-based list manipulation, and methods to remove items from list [#3329]

2.31 2026-06-17 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.h: update comments for methods about default_connect_timeout [#3329]
- tools/nut-scanner/scan_nut.c: nutscan_scan_ip_range_nut_authconf(): detect and load NUT auth conf file like other clients do [#3329]
- tools/nut-scanner/scan_nut.c: nutscan_scan_ip_range_nut_authconf(): detect desired timeout like other NUT clients do [#3329]

2.32 2026-06-16 Jim Klimov <jimklimov+nut@gmail.com>

- tools/nut-scanner/scan_nut.c: detect if authconf-related methods are present in libupsclient build we try to use [#3329]
- docs/man/nutscan_scan_nut.txt, tools/nut-scanner/Makefile.am, tools/nut-scanner/nut-scan.h, tools/nut-scanner/scan_nut.c: tools/nut-scanner/*, docs/man/nutscan_scan_nut.txt: introduce nutscan_scan_ip_range_nut_authconf() and nutscan_scan_nut_authconf() with nutscan_nut_authconf_t argument [#3329] Make old methods wrappers of the new ones, so existing libnutscan clients continue working as they were (no breaking ABI change, just new methods).
- scripts/obs/debian.rules, scripts/obs/nut.spec: update comments about SO_MAJOR_LIB* version hassle
- docs/nut.dict: sort the dictionary after PR #3447
- NEWS.adoc: mention PR #3448
- NEWS.adoc: mention PR #3456
- docs/Makefile.am: fix ASPELL_ENV_LC_ALL for platforms that care
- NEWS.adoc: refer to PR #3485

2.33 2026-06-15 Jim Klimov <jimklimov+nut@gmail.com>

- docs/ci-farm-do-setup.adoc, docs/nut.dict: add notes about ZFS and 4KB disk sectors

2.34 2026-06-14 mbv06 <mbv06.dev@gmail.com>

- NEWS.adoc, data/driver.list.in, drivers/nutdrv_qx.c: nutdrv_qx: Fix armac subdriver failing on 7-byte interrupt reads Certain UPS models (e.g. SVC VP-1250-LCD) return a 7-byte report instead of the standard 6 bytes. This relaxes the strict equality check in the interrupt read loop and increases the requested read size to 8 bytes to safely accommodate these devices without dropping trailing bytes or timing out.
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2.35 2026-06-12 Jim Klimov <jimklimov+nut@gmail.com>

- Jenkinsfile-dynamatrix: update optional traceable toggles

2.36 2026-06-11 mbv06 <mbv06.dev@gmail.com>

- NEWS.adoc: docs: note Windows state pipe security update

2.37 2026-06-11 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/richcomm_usb.c, NEWS.adoc: bump DRIVER_VERSION and document the improvements [#1768, #3477]

2.38 2026-06-10 dependabot[bot] <49699333+dependabot[bot]@users.noreply.github.com>

- .github/workflows/01-make-dist.yml: build(deps): bump bluwy/substitute-string-action from 3 to 4 Bumps [bluwy/substitute-string-action](#) from 3 to 4. - [Release notes](#) - [Commits](#) ---- updated-dependencies: - dependency-name: bluwy/substitute-string-action dependency-version: 4 dependency-type: direct:production update-type: version-update:semver-major ----
- .github/workflows/01-make-dist-worker.yml: build(deps): bump actions/upload-artifact from 4 to 7 Bumps [actions/upload-artifact](#) from 4 to 7. * [Release notes](#) * [Commits](#) ---- updated-dependencies: * dependency-name: actions/upload-artifact dependency-version: 7 dependency-type: direct:production update-type: version-update:semver-major ----

2.39 2026-06-09 mbv06 <mbv06.dev@gmail.com>

- docs/man/richcomm_usb.txt, drivers/richcomm_usb.c, tools/nut-scanner/scan_usb.c: Fix richcomm USB matching options
- drivers/richcomm_usb.c: drivers: preserve libusb context across USB reopens (richcomm only)

2.40 2026-06-08 mbv06 <mbv06.dev@gmail.com>

- drivers/dstate.c: drivers: fix Windows state pipe permissions

2.41 2026-06-08 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.h, clients/nutclient.cpp, clients/nutclient.h, scripts/perl/UPS/Nut.pm, scripts/python/module/PyNUT.py.in: clients/authconf.h, clients/nutclient.{h,cpp}, scripts/perl/UPS/Nut.pm, scripts/python/module/PyNUT.py.in: revise ability to parse nutauth.conf files [#3329]
- scripts/perl/UPS/Nut.pm: fix old trailing whitespaces

2.42 2026-06-07 Jim Klimov <jimklimov+nut@gmail.com>

- include/nutconf.hpp, common/nutconf.cpp: add ability to parse nutauth.conf files [#3329]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.h, scripts/perl/UPS/Nut.pm, scripts/python/module/PyNUT.py.in: clients/nutclient*.{h,cpp}, scripts/perl/UPS/Nut.pm, scripts/python/module/PyNUT.py.in: add ability to parse nutauth.conf files [#3329]

2.43 2026-06-06 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: upscli_authenticate(): reduce buf sizes for result to fit UPSCLI_NETBUF_LEN [#3329]
- clients/upsimage.c, clients/upsset.c, clients/upsstats.c: clients/ups*.c: comment in CGI clients about auto-login (not wise for that use case) [#3329, #3411]
- clients/upsc.c: support upscli_authenticate_authconf() [#3411, #3329]
- clients/upsclient.c, clients/upsclient.h, clients/upscmd.c, clients/upsmon.c, clients/upsrw.c, docs/man/Makefile.am, docs/man/upsc.txt, docs/man/upscli_authenticate.txt, docs/man/upscmd.txt, docs/man/upslog.txt, docs/man/upsrw.txt: clients/upsclient.{c,h}, docs/man, upscmd.c, upsrw.c, upsmon.c: refactor with upscli_authenticate() to unite different tricks [#3329] Introduce upscli_authenticate for completeness.
- docs/Makefile.am: fix SOURCE_CHECK_UNICODE for out-of-tree builds

2.44 2026-06-05 Jim Klimov <jimklimov+nut@gmail.com>

- clients/Makefile.am, configure.ac: configure.ac: revise LIBSSL_LIBS vs "-pthread" option (from NSS occasionally) for C+\+ builds [#3453] Address error: argument unused during compilation: *-pthread*
- Makefile.am, SECURITY.md, docs/Makefile.am: SECURITY.md: introduce a policy that can be seen on GitHub Describe the current situation with support of NUT versions and the parties involved, and how to responsibly disclose vulnerabilities.

2.45 2026-06-03 Jim Klimov <jimklimov+nut@gmail.com>

- .github/workflows/01-make-dist-worker.yml, .github/workflows/01-make-dist.yml, .github/workflows/05-codeql.yml, .github/workflows/PyNUTClient.yml: .github/workflows/*.yml: bump actions versions Originally proposed by dependabot, but we cannot put its Markdown commit comments into ChangeLog.pdf, so reposting.
- .github/dependabot.yml: introduce scanning for obsolete GHA versions
- .github/workflows/01-make-dist.yml: update comments about the Checks message [#1400]
- .github/workflows/01-make-dist.yml: when dispatching the workflow for a branch iteration, use its name (ref) not commit [#1400]
- .github/workflows/01-make-dist.yml: delete old artifacts OTHER than the one just created [#1400] Before the recent refactoring, the logic was to first build the tarball, then if it succeeded - clean up the storage from obsolete ones, and quickly publish the new one. Now that a separate workflow does the uploads, we can at best ensure deletion of obsolete tarballs after hopefully having the new one up there (had enough space and all that). Also take care to not delete the newly made artifact ;)
- NEWS.adoc: update for FreeBSD and Jenkins bumps
- NEWS.adoc: Rearranged the GHA job to prepare Dist and Docs Tarballs [#1400, #3460]
- .github/workflows/01-make-dist.yml: launch "GHA-01: Tarballs Build Worker" definition from the target branch (if a PR) [#1400]
- configure.ac: pass -Wno-unused-command-line-argument also to CLANG LDFLAGS
- .github/workflows/01-make-dist.yml: DRY: define artifact_name once [#1400]

2.46 2026-06-02 Jim Klimov <jimklimov+nut@gmail.com>

- .github/workflows/01-make-dist-worker.yml, .github/workflows/01-make-dist.yml: .github/workflows/01-make-dist{,-worker}.yml: retell the story of two workflows [#1400]
 - .github/workflows/01-make-dist.yml: reduce the permissions needed for artifact manipulation [#1400]
-

2.47 2026-06-01 Jim Klimov <jimklimov+nut@gmail.com>

- Jenkinsfile-dynamatrix: use new countStagesPerNode feature from the library in notifyHandlerAlert
- Jenkinsfile-dynamatrix: introduce dynacfgPipeline.notifyHandlerAlert (to report looping RESTART dynamatrix cells)

2.48 2026-05-31 Jim Klimov <jimklimov+nut@gmail.com>

- configure.ac: ignore recent clang warning about unused-command-line-argument

2.49 2026-05-31 Doug Nazar <nazard@nazar.ca>

- common/common.c, drivers/upsdrvquery.c, include/common.h, server/sstate.c: Use non-blocking connect() for AF_UNIX sockets to ensure we don't stall. If a driver has become unresponsive, connect() will block after the listen() backlog has been exhausted, causing the main upsd to also become unresponsive.

2.50 2026-05-28 Jim Klimov <jimklimov+nut@gmail.com>

- tools/check-source-nonascii.pl: fix syntax for hash length
- ci_build.sh: fix normalize_path_perl() syntax for hash length; avoid newlines

2.51 2026-05-27 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt: document JDK21+ as the requirement for NUT CI build farm agents since spring 2026
 - ci_build.sh: normalize_path() before reporting it at start of actual work about BUILD_TYPE
 - ci_build.sh: configure_nut(): report the PATH value ultimately used (maybe normalized)
 - ci_build.sh: propose shell vs perl (faster?) implementations of normalize_path()
 - drivers/bicker_ser.c, drivers/nut-libfreeipmi.c, drivers/phoenixcontact_modbus.c, drivers/socomec_jbus.c, drivers/usbhid-ups.c: drivers/*.c: replace Unicode in comments with ASCII equivalents
 - drivers/socomec_jbus.c: clean up trailing whitespace
 - docs/documentation.txt, docs/man/nut.txt: replace Unicode long dashes with acsii --
 - drivers/libwinhid.h: replace Unicode long dashes with acsii -- and arrow with ==>
 - docs/developers.txt, docs/nut.dict: docs/developers.txt: add a chapter on ASCII vs Unicode
 - Makefile.am, configure.ac, docs/Makefile.am, tools/Makefile.am, tools/check-source-nonascii.pl: configure.ac, Makefile.am, docs/Makefile.am, tools/check-source-nonascii.pl: add unicode-in-sources detection to spellcheck. On some platforms, at least the file type parsing fails for text with some but not other wide characters (so it is "data" and not some sort of "text"). This test was added for C sources as well, but keep in mind that the final authority is the actually used compilers - which are free to complain or not.
 - .github/pull_request_template.md: update with a section on Unicode characters
 - configure.ac: report the tested compiler location just before AC_PROG_CC etc. which might fail
 - configure.ac: unset CCACHE_DIR if provided but empty. Notably, ccache-3.7.12 in FreeBSD 15 breaks CI build with that complaint.
 - drivers/ragtech.c: pre-declare variables for portability [#3447]
 - drivers/ragtech.c: replace Unicode long dashes with acsii -- [#3447]
 - docs/man/ragtech.txt: replace Unicode long dashes with acsii -- [#3447]
-

2.52 2026-05-26 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt, docs/nut.dict: revise with FreeBSD 15 lessons learned
- configure.ac: if developer enables configure debug, only wipe DOCTESTDIR after confirming we can build docs we want
- drivers/ragtech.c: fix compiler warnings about %u vs. conditionally picked numbers [#3447] ... which are signed int by default

2.53 2026-05-25 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc, UPGRADING.adoc, docs/man/upsd.conf.txt, server/netssl.c: server/netssl.c: ssl_init(): abort with fatalx() and better explanations when SSL setup was requested but failed [#3331]
- tests/NIT/nit.sh: check_NIT_certs_NSS(): fix the check for *.txt and *.db file patterns [#3331, #1711]
- tests/NIT/nit.sh: wrap long openssl lines; fix a missed validity-time spec [#1711]
- tests/NIT/nit.sh: check_NIT_certs_NSS(): verify old run-time toolkit vs. newer (cached) DB files [#3331, #1711] Auto-repair the trusted NSS DB set(s) of files if we can. Primarily relevant for build agents where same cache is used by NSS toolkits a decade apart.
- docs/nut.dict: add opcodes
- configure.ac: add -Wno-thread-safety-negative to clang flags (seen on FreeBSD builds)

2.54 2026-05-24 Pedro Cunha <pedroagracio+nut@gmail.com>

- drivers/libusb1.c: use explicit context; drop per-close libusb_exit() [#598] The original code called libusb_init(NULL) on every nut_libusb_open() and libusb_exit(NULL) on every nut_libusb_close(), bumping and decrementing the default context's refcount. That violates the documented contract for libusb_exit, which "should be called after closing all open devices and before your application terminates", and on certain firmware it wedges indefinitely: when libusb_exit hits the 1→0 transition it tears the context down, which waits on libusb's internal sync primitives for outstanding URBs to drain. URBs orphaned by libusb_reset_device or by an unexpected device disconnect never drain, so the wait never returns. The deadlock is reachable from any reconnect path. It was first recognized by 4f84b7f92 ("don't libusb_exit() when closing a previously opened device"), which dropped the call in nut_libusb_open()'s rematch loop. The fallback reconnect path in qx_command and the rest of the driver lifecycle still hit it. With the companion change in nutdrv_qx.c (a4813f868) that escalates persistent LIBUSB_ERROR_OVERFLOW to libusb_reset_device, the deadlock window opens on every escalation and was hit reproducibly on the 0665:5161 Cypress USB-serial bridge family (Salicru SPS, Ippon, ViewPower, Voltronic Power UPSes; see [#598, #993, #2453]). Switch to an explicit libusb context owned end-to-end: - Initialize once on the first nut_libusb_open() and register a matching one-shot atexit handler. - Remove libusb_exit() from nut_libusb_close(); the close path now only calls libusb_close(), which is non-blocking. - Replace libusb_get_device_list(NULL, ...) with the explicit context so the default context is never touched.
- Bound the atexit libusb_exit() with a SIGALRM-based 2s timeout (POSIX) so a deadlock-prone teardown at shutdown can't wedge supervisor stop sequences (e.g. systemctl stop). Sub-millisecond on a healthy context. Bump USB_DRIVER_VERSION from 0.53 to 0.54. Validated on a Salicru SPS 1500 ONE BL (0665:5161) by authorize-toggle stress reproduction (8 cycles, clean reconnect each time, same driver PID throughout) and by a natural OVERFLOW event captured in soak (counter-gated logic at 1/3, next poll succeeded, no escalation; driver PID stable for 22h+).

2.55 2026-05-22 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c, clients/authconf.h, docs/man/upscli_dump_authconf_item.txt: clients/authconf.{c,h}, docs/man/upscli_dump_authconf_item.txt: do not specify "FILE *restrict stream", some platforms think "restrict" is the variable name [#3329]
 - lib/libupsclient-config.in: this should consider LIBSSL_LDFLAGS_RPATH [#3331]
 - lib/libnutclient.pc.in: this now Requires: @LIBSSL_REQUIRES@ [#3331]
-

2.56 2026-05-21 Jim Klimov <jimklimov+nut@gmail.com>

- tests/Makefile.am: test_authconf: consider LIBSSL flags [#3329, #1711] Builds with NSS tend to fail on some platforms due to not locating libnss3.so (which is hidden from common searches in an nss/mps subdirectory)

2.57 2026-05-20 Pedro Cunha <pedroagracio+nut@gmail.com>

- drivers/nutdrv_qx.c: qx_command(): reset USB device after persistent LIBUSB_ERROR_OVERFLOW [#598] LIBUSB_ERROR_OVERFLOW has been handled as a benign, retry-on-next-poll condition (grouped with LIBUSB_ERROR_TIMEOUT and the default case) ever since the original blazer import. For most devices a one-off oversized interrupt-IN frame is indeed transient. But some Cypress USB-serial bridge firmware (VID:PID 0665:5161; Salicru SPS, Ippon, ViewPower and various Voltronic Power UPSes) wedges the endpoint once it overruns: every subsequent interrupt read returns OVERFLOW and the driver spins in the stale-data loop until an external USB-level reset. This is the chronic hang reported in [issue #598](#). Distinguish the two cases with a small consecutive-overflow counter: the first QX_USB_OVERFLOW_RESET_TRIES-1 overflows are retried on the next poll (so genuine transients cost nothing but a skipped cycle), and a sustained run escalates to usb_reset() \+ reconnect, reusing the recovery path already used for PIPE/ETIME/IO errors. Any clean read resets the counter. Captured on a Salicru SPS 1500 ONE BL: an isolated overflow recovers on the next poll, while a sustained lockup (dozens of consecutive overflows within seconds) is cleared only by the device reset, matching the behaviour of an external `usb_resetter --reset-device`.

2.58 2026-05-20 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsstats.c: ups_connect(): avoid potential NULL dereference in debug/failure printouts

2.59 2026-05-19 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc, clients/upsc.c, clients/upscmd.c, clients/upsimage.c, clients/upslog.c, clients/upsrw.c, clients/upsset.c, clients/upsstats.c, docs/man/upsc.txt, docs/man/upscmd.txt, docs/man/upsimage.cgi.txt, docs/man/upslog.txt, docs/man/upsrw.txt, docs/man/upsset.cgi.txt, docs/man/upsset.conf.txt, docs/man/upsstats.cgi.txt: clients/{upscmd,upsrw,upslog,upsstats,upsset,upsimage}.c, docs/man/*.txt: add support for nutauth.conf [#3329]
- clients/upsclient.c: upscli_cleanup(): call upscli_free_authconf_list() [#3329]
- clients/upsclient.c, docs/man/upscli_add_host_cert.txt: refactor upscli_find_host_port_cert() with a "verbose" option, and extend upscli_add_host_port_cert() with check for existing entries [#3329]
- clients/authconf.c, drivers/libusb0.c, include/Makefile.am, include/strcasestr-static.h: include/strcasestr-static.h, clients/authconf.c, etc.: promote code from drivers/libusb0.c to be a bit more shared [#3329] Follows up from commits b91e34edc79 and f951dce8deb
- clients/upsclient.c: upscli_find_host_cert(), upscli_add_host_cert(): avoid "s_host" varname which confuses WIN32, illumos and other builds [#3329]
- clients/authconf.c, docs/man/nutauth.conf.txt, tests/NIT/nit.sh: set_authconf_val(): support USERNAME as alias of USER [#3329] Follow the NUT Networked Protocol keywords a bit more closely.
- tests/NIT/nit.sh: do not export NUT_AUTHCONF_FILE to NIT.env [#3329, #1711] Depending on context, whether "none" or a real path may be useful or toxic to custom developer test works.
- clients/upsclient.c: AuthCertificate(), upscli_sslinit(): refactor NSS server certname validation [#3331] It is NOT about just host name (URL) matching per our spec.
- clients/upsclient.c: revise logging from AuthCertificate*() methods [#3329]
- tests/NIT/nit.sh: generatecfg_upsmon_add_SSL(): consider CERTHOST with non-default NUT_PORT now [#3329]
- conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: clarify CERTHOST with non-default NUT_PORT [#3329]
- clients/upsclient.c: upscli_add_host_cert(): fix reporting for plain legacy "hostname" input [#3329]

- clients/upsclient.c: BadCertHandler(): cast printing of port [#3329]
- clients/upsclient.c: upscli_sslinit(): wrap long lines in NSS part [#3329]
- tests/NIT/Makefile.am: check-NIT-devel: minimize nut_version.h regeneration attempts [#1711]
- tests/NIT/nit.sh: fix passing server cert trust to NSS DB [#3331, #1711]
- tests/NIT/nit.sh: fix passing server cert usage to NSS DB [#3331, #1711]
- tests/NIT/nit.sh: fix passing root CA cert trust to NSS DB [#3331, #1711]

2.60 2026-05-18 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apc_common.c, drivers/apc_common.h: add Axel Gembe copyright to apc_common as the code is at the very least inspired by the original code from apc-modbus

2.61 2026-05-18 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: upscli_find_host_port_cert(): debug-trace whether this succeeded [#3329]
- clients/upsclient.c: upscli_add_host_cert(): revise splitting apart the "hostname" which may come from authconf section name [#3329]
- clients/upsclient.c: upscli_init2(): quiesce reaction to NUT_QUIET_INIT_SSL=false [#1711]
- clients/upsclient.c: upscli_init_authconf(): if the "ac" refers to a CERTHOST, call upscli_add_host_cert() on it [#3329]
- clients/upsclient.c: upscli_init_authconf(): debug-trace the authconf pointer [#3329, #1711]
- clients/authconf.c, clients/authconf.h, docs/man/upscli_dump_authconf_item.txt, tests/test_authconf.c: clients/authconf.{c,h}, docs/man/upscli_dump_authconf_item.txt, tests/test_authconf.c: extend upscli_dump_authconf_{list,item}() with an option to show/hide passwords [#3329]
- NEWS.adoc, UPGRADING.adoc, clients/upsc.c, docs/man/upsc.txt: clients/upsc.c, docs: add support for "-A /path/to/nutauth.conf" [#3329, #3411]
- clients/upsclient.c: document better the upscli_init*() and upscli_sslinit() methods [#3329]
- clients/authconf.c: handle_authconf_args(): revise logging of parsed lines, especially malformed and sensitive ones [#3329]
- tests/test_authconf.c: consult presence of NUT_AUTHCONF_FILE envvar to parse and print out that file [#3329, #1711]
- tests/NIT/nit.sh: add support for nutauth.conf file testing, add a read-only user [#3329, #1711, #3411] Call new generatecfg_nutauth() after we do generatecfg_upsdusers_trivial() (and "coincidentally" in one case generatecfg_upsd_add_SSL() as well).
- clients/upsclient.c: upscli_add_host_cert(), upscli_find_host_cert(): refactor with shared get_port_from_string() logic [#3329]
- drivers/apcmicrolink.c: drop pragma to ignore a warning unknown to old clang [#3406]

2.62 2026-05-17 juslex <66561713+juslex@users.noreply.github.com>

- NEWS.adoc, data/driver.list.in, docs/man/Makefile.am, docs/man/ragtech.txt, docs/nut.dict, drivers/Makefile.am, drivers/ragtech.c, include/wincompat.h: ragtech: new driver for Ragtech UPSes (USB CDC-ACM, family 10) Targets Brazilian-built Ragtech "Easy Pro" / NEP / TORO / INNERGIE / OneUP devices that present themselves as USB CDC-ACM (VID 0x04D8, PID 0x000A, Microchip PIC firmware). Validated end-to-end against an Easy 2000 TI (reg 0x9A model id = 16) read out of a working OneUP Nitro 2000. Protocol — three opcodes observed in OEM traffic: 0x01 ADDR_HI ADDR_LO VALUE CKSUM write byte 0x02 ADDR_HI ADDR_LO MASK CKSUM AND mask (atomic bit-clear) 0x04 ADDR_HI ADDR_LO COUNT CKSUM read range CKSUM = (ADDR_HI \+ ADDR_LO + VALUE) & 0xFF The CDC-ACM channel ignores baud

at the wire but DTR/RTS are interpreted by some Ragtech families as a remote shutdown signal — the driver forces both low after open and does NOT call `ser_set_speed()` to avoid the `tcsetattr()` DTR pulse that some Linux tty drivers perform. The full 30-byte main range (0x80..0x9D) plus `V_IOUTCALIB` (0xF3) and the oscillator calibration pair (0x202/0x203) are read; output frequency is interpolated per `devices.xml` formula rather than hard-coded. Twenty models from the family-10 device table populate `ups.model`, `ups.realpower.nominal`, `output.voltage` scaling, `output.current` scaling and battery voltage scaling. `input.voltage.nominal` switches between 115V and 220V dynamically based on the measured input. Instcmds implemented: `shutdown.stayoff` — aa 02 00 80 fe (clear AUTOSTART) \+ aa 01 00 98

- A. Byte-for-byte identical to the OEM `supsvc` sequence. Validated: cuts output and stays off until manual power-on, in both AC and battery modes. `shutdown.return` — falls back to `shutdown.stayoff` with a warning. The atomic OR opcode that would set `F_AUTOSTART` has not been captured yet; read-modify-write with 0x01 sends the bytes but does not trip the firmware's shutdown state machine. `shutdown.stop` — aa 01 00 98 00 (write `V_SHUTDOWNTIMER` = 0). Confirmed to abort an armed countdown. `test.battery.start.deep` — bit writes on regs 0x90 / 0x95 (`fullDischarge` in `devices.xml`). Flagged as unverified for the same reason as `shutdown.return`. Read primitive uses `ser_get_buf()` in a manual loop rather than `ser_get_buf_len()` (which discards partial reads on timeout — CDC fragments replies into 1-8 byte chunks). `ser_flush_io()` is used before each TX to clear the kernel CDC buffer (`ser_flush_in()` only does a select+read loop that misses queued bytes under `O_NONBLOCK`). Reverse engineering primarily from the OEM `devices.xml` table for register layout / scaling / flags / actions, and from trace of the OEM `supsvc` binary capturing both read polls and the LED/shutdown write sequences. Cross-checked against `UPS_ESP32_tinySrv` (https://github.com/antunesls/UPS_ESP32_tinySrv).

2.63 2026-05-15 Lukas Schmid <lukas.schmid@netcube.li>

- docs/man/apcmicrolink.txt, docs/nut.dict, drivers/apcmicrolink.c: update docs and spellcheck dictionary

2.64 2026-05-15 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c, docs/man/upscli_read_authconf_file.txt: add support for `NUT_AUTHCONF_FILE` and/or `NUT_AUTHCONF_PATH` envvars to locate exactly one `nutauth.conf` candidate or bail out [#3329]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: add rudimentary support for `SSLConfig_CERTHOST` with a port number [#3329]
- clients/authconf.c, clients/upsclient.c, clients/upsclient.h, docs/man/Makefile.am, docs/man/upscli_add_host_cert.txt: clients/upsclient.{c,h}, clients/authconf.c, docs: introduce `upscli_add_host_port_cert()` et al [#3329]
- clients/authconf.c, tests/test_authconf.c: `handle_authconf_args()`: when we finish parsing a section, `upscli_add_host_cert()` if applicable [#3329]
- clients/authconf.c, clients/authconf.h, docs/man/nutauth.conf.txt, docs/man/upscli_create_authconf_item.txt: clients/authconf.{c,h}, docs/man/upscli_create_authconf_item.txt, docs/man/nutauth.conf.txt: add `CERTHOST` field support [#3329]

2.65 2026-05-14 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/Makefile.am: add missing `apc_common.h` to distributed headers
- drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: extract descriptor helpers and fix map includes

2.66 2026-05-14 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c: `upscli_get_authconf_item()`: add ifdef-ed away very optional debug messages [#3329]
- tests/test_authconf.c: print TAP-style progress tracking| add tests for `upscli_get_authconf_item()` modification of the list [#3329]
- clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/upscli_find_authconf_item.txt, tests/test_authconf.c: clients/authconf.{c,h}: introduce `upscli_get_authconf_item()` for auto-merged items [#3329]

- docs/man/Makefile.am: fix after reshuffle nutauth.conf related methods vs population of man pages [#3329]
- clients/authconf.c, docs/man/upscli_create_authconf_item.txt: revise upscli_merge_authconf_item() function vs. cloning of empty strings [#3329]
- clients/authconf.c, docs/man/upscli_create_authconf_item.txt: revise upscli_clone_authconf_item() and upscli_merge_authconf_item() functions vs. cloning of USER name [#3329]
- clients/authconf.c: refactor with separation of upscli_add_authconf_item(name) and upscli_add_authconf(node) [#3329]
- clients/authconf.c: upscli_find_authconf_item(): refine return of global_defaults and zsearch in empty list [#3329]

2.67 2026-05-13 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apcmicrolink.c, drivers/apcmicrolink.h: fix switch warnings and replace flag bit-fields

2.68 2026-05-13 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/upscli_create_authconf_item.txt, docs/man/upscli_get_authconf_item.txt, docs/man/upscli_dump_authconf_item.txt, docs/man/upscli_find_authconf_item.txt, docs/man/upscli_free_authconf_item.txt, docs/man/upscli_get_authconf_list.txt, docs/man/upscli_init.txt, docs/man/upscli_read_authconf_file.txt, tests/test_authconf.c, clients/authconf.{c,h}, docs: introduce upscli_clone_authconf_item() and upscli_merge_authconf_item() functions [#3329]
- clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/nutauth.conf.txt, ..._authconf.txt ⇒ upscli_create_authconf_item.txt, ..._authconf.txt ⇒ upscli_dump_authconf_item.txt, ..._authconf.txt ⇒ upscli_find_authconf_item.txt, docs/man/upscli_free_authconf_item.txt, docs/man/upscli_get_authconf_list.txt, docs/man/upscli_init.txt, ..._authconf.txt ⇒ upscli_read_authconf_file.txt, tests/test_authconf.c, clients/authconf.{c,h}, tests/test_authconf.c, docs/man/auth, docs/man/Makefile.am: rename file/list/item/section methods consistently with their purpose [#3329]
- clients/authconf.c: upscli_create_authconf(): handle failed calloc() by returning NULL; fail fatally in upscli_add_authconf() if needed [#3329]
- docs/man/Makefile.am, docs/man/upscli_create_authconf.txt, ..._authconf_list.txt ⇒ upscli_dump_authconf.txt, docs/man/upscli_find_authconf.txt, docs/man/upscli_free_authconf_list.txt, docs/man/upscli_get_authconf_list.txt: docs/man/auth.txt, docs/man/Makefile.am: reshuffle nutauth.conf related methods vs population of man pages [#3329]
- clients/authconf.c, docs/man/nutauth.conf.txt: revise section title parsing, INCLUDE handling, naming of [_global_defaults], code comments etc. [#3329]
- clients/authconf.c, clients/authconf.h: clients/authconf.{c,h}: refactor static upscli_add_authconf() with a public upscli_create_authconf() [#3329]
- clients/authconf.c: fix static analysis warnings [#3329]
- tests/test_authconf.c: add a Non-exact match test case [#3329]
- clients/authconf.c: refactor upscli_find_authconf() with upscli_normalize_auth_section_parts() [#3329]
- NEWS.adoc, drivers/nutdrv_qx_megatec.c: drivers/nutdrv_qx_megatec.c: add QX_FLAG_ABSENT to "ups.firmware" so devices with all-spaces in that field can still be recognized [#3436] Co-authored-by: Jim Klimov <jimklimov+nut@gmail.com> Co-authored-by: Siravijbb <siravijbb@gmail.com>

2.69 2026-05-12 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apc_modbus.c: remove unused function
-

2.70 2026-05-12 Jim Klimov <jimklimov+nut@gmail.com>

- clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/upscli_find_authconf.txt: clients/authconf.{c,h}, docs: further refactor with upscli_normalize_auth_section_parts() which we can share and expose [#3329]
- clients/authconf.c: handle_authconf_args(): revise section title line parsing [#3329]
- clients/authconf.c: upscli_splitauth(): reject empty non-NULL sect_name [#3329]
- clients/authconf.c: upscli_splitauth(): if "port" is a non-numeric string, try to resolve it in the system naming database [#3329]
- scripts/obs/debian.nut-client.install, scripts/obs/debian.nut-client.manpages, scripts/obs/nut.spec: know about nutauth.conf samples [#3329]
- conf/nutauth.conf.sample.in, configure.ac, conf/Makefile.am, conf/.gitignore: add example config for nutauth.conf [#3329]
- tests/test_authconf.c: add a dev-test for discovery of user/site nutauth configs [#3329]
- tests/test_authconf.c: honour NUT_DEBUG_LEVEL [#3329]
- clients/authconf.c, clients/authconf.h, docs/man/upscli_read_authconf.txt: clients/authconf.{c,h}, docs/man/upscli_read_authconf.txt: add searching for default nutauth.conf in user home or NUT_CONFPATH locations [#3329]
- clients/upsclient.c, clients/upsclient.h, docs/man/Makefile.am, docs/man/upscli_init.txt: clients/upsclient.{c,h}, docs/man/upscli_init.txt: introduce upscli_init_authconf() [#3329]
- clients/authconf.c: parse_authconf_file(): call check_perms(filename) [#3329, #3359]
- common/common.c, include/common.h, server/upsd.c, server/upsd.h, server/user.h: common/common.c et al: relocate check_perms() from upsd to common code base [#3359]
- clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/upscli_find_authconf.txt, docs/nut.dict: clients/authconf.{c,h}, docs/man/upscli_find_authconf.txt: refactor with upscli_split_auth_section() [#3329]
- clients/authconf.c, clients/authconf.h, docs/man/upscli_dump_authconf_list.txt, tests/test_authconf.c: clients/authconf.{c,h}, tests, docs: extend upscli_dump_authconf{,_list} methods with "for_debug" option [#3329]
- tests/test_authconf.c: drop "Mocks for functions usually provided by libcommon" [#3329]
- common/common.c: cosmetic fix
- docs/Makefile.am: add ASPELL_NUT_COMMON_ARGS to preclude mangling of nut.dict on some platforms when checking interactively

2.71 2026-05-12 Doug Nazar <nazard@nazar.ca>

- common/common.c, drivers/main.c, include/common.h: drivers/main.c: Split backgrounding driver into two stages This is needed in particular with libusb on linux as it will create a thread to handle either netlink or udev messages. If this is created before we background, the driver will hang during exit trying to join the non-existent thread in the new process. Shows this warning while stopping the driver. Stopping /run/nut/usbhid-ups-xxxx.pid failed, retrying harder: Success

2.72 2026-05-11 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apcmicrolink.c: fix commands and writes sometimes not working and allow configuration of command source
- drivers/Makefile.am, drivers/apc_common.c, drivers/apc_common.h, drivers/apc_modbus.c, drivers/apc_modbus.h, drivers/apcmicrolink.c, drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: Variable cleanup and commons creation between apc-modbus and microlink

2.73 2026-05-11 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc, clients/Makefile.am, clients/authconf.c, clients/authconf.h, docs/man/Makefile.am, docs/man/nutauth.conf.txt, docs/man/upscli_dump_authconf_list.txt, docs/man/upscli_find_authconf.txt, docs/man/upscli_get_authconf_list.txt, docs/man/upscli_read_authconf.txt, docs/nut.dict, tests/Makefile.am, tests/test_authconf.c: Introduce NUT "authconf" file support [#3329] First PoC from AI, slightly modified in review, following the spec requested in the GitHub issue. Stepping stone for further work.
 - server/upsd.c: update_sysmaxconn(): default the sysmaxconn value to "MAX - RESERVE_FD_COUNT_UPSD" on platforms where it might matter [#3365]
 - server/upsd.c: update_sysmaxconn(): check that detected sysmaxconn_hard is big enough to be useful considering the RESERVE_FD_COUNT_UPSD set aside AND a few actual connections [#3365]
 - server/upsd.c: update_sysmaxconn(): revise warning message about hard limit [#3365]
 - server/upsd.c: be sure to re-detect sysmaxconn after (re-)loading config [#3365]
 - NEWS.adoc, UPGRADING.adoc, docs/nut.dict, server/upsd.c, server/upsd.h: server/upsd.{c,h}, NEWS.adoc, UPGRADING.adoc: consider getrlimit() permissions vs. MAXCONN request [#3365]
 - NEWS.adoc, server/conf.c, server/upsd.c, server/upsd.h: server/conf.c, server/upsd.{c,h}, NEWS.adoc: if we hit "Too many open files" during configuration reload, close oldest client connection and retry [#3365]
 - server/upsd.h: update © heading and whitespace styling
 - docs/nut.dict: update for dllldd [#3420]
 - NEWS.adoc: update about scripts/Windows/dllldd.sh helper script refinements [#3420]
 - docs/man/upscli_add_host_cert.txt: update that this call works also with OpenSSL (since NUT v2.8.6) [#3331]
 - docs/download.txt, docs/nut.dict: Upload NUT-for-Windows-x86_64-RELEASE-2.8.5-1-fixNSS.7z with NSS dependency libraries missing in original NUT v2.8.5 release [#3420]
 - appveyor.yml: allow ln to create hard links, and tell 7-zip to store them as such, not as independent files (subject to 7zip format du jour)
 - scripts/Windows/dllldd.sh: Revise use of TEMPFILE_REC to avoid grep if it is empty [#3420]
 - Makefile.am: install-win-bundle-thirdparty: revise progress message markup for easier troubleshooting [#3420]
 - scripts/Windows/dllldd.sh: dllldd_with_strings(): filter away OUT_TOOLS hits if known [#3420]
 - scripts/Windows/dllldd.sh: dllldd_with_strings(): quick bail-out with TEMPFILE_REC [#3420]
 - scripts/Windows/dllldd.sh: revise cleanup of TEMPFILE_REC; create it earlier in dllldddir() [#3420]
 - scripts/Windows/dllldd.sh: dllldd(): use TEMPFILE_REC if defined by caller [#3420]
 - scripts/Windows/dllldd.sh: dllldd_with_tools(), dllldd(): optimize looped iteration of search paths we prepared [#3420] Avoid calls to `tr ' : ' '\n'` that can be done once.
 - scripts/Windows/dllldd.sh: relocate TEMPFILE_REC declaration to before first use-mention [#3420]
 - scripts/Windows/dllldd.sh: filter_away_system_DLLs(): add some more OS-provided common hits to exclusion list [#3420]
 - scripts/Windows/dllldd.sh: filter_away_system_DLLs(): add DWrite.dll [#3420] Some OS releases have Microsoft DirectWrite for high-quality text rendering, anti-aliasing, and font management. . . not something we really care about in NUT, and part of OS footprint we do not drill into.
 - scripts/Windows/dllldd.sh: dllldd_with_tools(), dllldd(): optimize SEARCH_INPUT_PATH discovery [#3420] Instead of a loop with `dirname` calls (spawning those processes takes a few seconds in MSYS2 on Windows), use one `sed` process.
 - scripts/Windows/dllldd.sh: dllldd_with_strings(): ignore also "lib%s.dll" hits [#3420]
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- scripts/Windows/dllldd.sh: accept also outputs from "objdump" variants that return "NEEDED libname" [#3420]
- scripts/Windows/dllldd.sh: when given a CLI argument, check if it is a directory (to run dlllddir) before diving in with dlllddrec [#3420]
- scripts/Windows/dllldd.sh: accept also outputs from "ldd" variants that only return path (no parenthesis with hex) [#3420]
- scripts/Windows/dllldd.sh: allow to parameterize DLLEXTR_REGEX [#3420] At least, allow to test/iterate the script with non-Windows builds too.
- scripts/Windows/dllldd.sh: dllldd_with_strings(): filter away some bogus strings outputs and findings [#3420]
- scripts/Windows/dllldd.sh: more filter_away_system_DLLs() [#3420]

2.74 2026-05-10 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/Windows/dllldd.sh: dllldd_with_tools(): define and clean-up TEMPFILE_REC in the outer-most call if we stack them [#3420] This should help directory crawlers etc. ignore paths already walked by some previous recursion in this scripted execution.
- scripts/Windows/dllldd.sh: dllldd_with_tools(): do not bail out if we found some but not all DLLs via objdump; feed only the names we did not resolve to ldd [#3420]
- scripts/Windows/dllldd.sh: consider also SEARCH_INPUT_PATH based on where the input EXE/DLL files are [#3420] This allows to find libnutprivate*.dll or libupsclient*.dll files we built and installed to EXE bindir, which were otherwise ignored by earlier path discovery tools and/or logic.
- scripts/Windows/dllldd.sh: filter_away_NUT_DLLs() when we query both dllldd_with_tools() and dllldd_with_strings() [#3420]
- scripts/Windows/dllldd.sh: refactor with discover_COMPILER_PATHS() to not do this in a loop [#3420]

2.75 2026-05-09 Jim Klimov <jimklimov+nut@gmail.com>

- docs/nut.dict: Update nut.dict [#3430]

2.76 2026-05-08 ostrich <570911+ostrich@users.noreply.github.com>

- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: Fix NUT-Monitor tray tooltip text

2.77 2026-05-08 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: update about NUT-Monitor tooltips [#3430]

2.78 2026-05-07 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsmon.c, NEWS.adoc: do not abort if we did not succeed upscli_init() but did not require SSL in the first place [#3420]
- scripts/Windows/dllldd.sh: refactor to find DLLs with strings too; track filenames we already investigated [#3420] DRY the loop and avoid cycles.

2.79 2026-05-06 Jim Klimov <jimklimov+nut@gmail.com>

- m4/nut_check_libnetnmp.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - m4/nut_check_libavahi.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - ci_build.sh, scripts/Windows/build-mingw-nut.sh: get_CI_CACHE_NUT_HASHDIR_CFG_OPT(): consider also MAKE and (CONFIG_)SHELL as circumstances that impact the configuration hash [#3108]
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2.80 2026-05-05 Jim Klimov <jimklimov+nut@gmail.com>

- m4/nut_check_libfreeipmi.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libgd.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libglib.m4, configure.ac: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libgpod.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libmodbus.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libneon.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libltdl.m4: do not drag around nut_have_libltdl in cached section, use nut_cv_have_libltdl right away [#3108]
- m4/nut_check_libopenssl.m4, m4/nut_check_libnss.m4, configure.ac: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libpowerman.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_libusb.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- m4/nut_check_os.m4: fix comments starting a line with a hash to use "dnl" first

2.81 2026-05-04 Jim Klimov <jimklimov+nut@gmail.com>

- m4/nut_check_libregex.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - m4/nut_check_libsystemd.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - m4/nut_check_libwrap.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - m4/nut_check_pkgconfig.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
 - m4/nut_check_os.m4: set OS_NAME and PKG_TARGET after passing AC_CACHE_CHECK [#3108]
 - m4/nut_check_libltdl.m4: do not reset nut_have_libltdl=yes when checking for header from cached data [#3108]
 - m4/nut_check_libltdl.m4: add a summary print-out for cached re-runs [#3108]
 - common/common.c: upsnotify(): fix "unused" warnings in different build combos
 - ci_build.sh, ci_build.adoc: when we DO_USE_NUTCI_CACHE_DEBUG, stash also the help text from configure script [#3108]
 - m4/nut_check_pkgconfig.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libwrap.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libusb.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libsystemd.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libregex.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libpowerman.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libopenssl.m4, m4/nut_check_libpowerman.m4: m4/nut_check_libopenssl.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libnss.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libnetsnmp.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
 - m4/nut_check_libneon.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
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- m4/nut_check_libmodbus.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
- m4/nut_check_libgpiod.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
- m4/nut_check_libglib.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
- m4/nut_check_libgdl.m4: fix indentations, use AS_IF and AS_CASE more [#3108]
- m4/nut_check_libfreeipmi.m4: fix indentations, use AS_IF [#3108]
- m4/nut_check_libavahi.m4: fix indentations, use AS_IF [#3108]
- m4/nut_check_cppcheck.m4: fix indentations, use AS_IF and AC_CACHE_VAL [#3108]
- m4/nut_check_bool.m4: fix indentations [#3108]
- m4/nut_check_aspell.m4: fix indentations, use AS_IF [#3108]
- m4/nut_check_asciidoc.m4: fix indentations, use AS_IF [#3108]
- m4/nut_check_libltdl.m4: fix indentations, use AS_IF for outer check [#3108]
- m4/nut_check_os.m4: fix typo in heading
- m4/nut_check_libltdl.m4: revise with AC_CACHE_VAL() for each option to be remembered [#3108]
- Makefile.am: when we touch -d "-10 seconds", hide the errors (such request is known to be not too portable)
- ci_build.sh, scripts/Windows/build-mingw-nut.sh: facilitate re-runs with changed **.m4/.am/.ac sources and enabled autoconf cache: handle precious CCACHE_ args/vars** [#3108]
- ci_build.adoc: clarify some points about caching [#3108]

2.82 2026-05-02 Jim Klimov <jimklimov+nut@gmail.com>

- ci_build.sh, ci_build.adoc: introduce DO_USE_AUTOCONF_CACHE_DEBUG=each mode [#3108]

2.83 2026-05-01 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: move entry for #3302 into v2.8.6 release

2.84 2026-04-30 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt: do not pull valgrind into default Debian/Ubuntu footprint, it is not ubiquitous Revise other suggestions a bit too

2.85 2026-04-29 Jim Klimov <jimklimov+nut@gmail.com>

- tests/cpptest-client.cpp: revise reporting of UPSCLI_SSL_CAPS* [#3331, #1711]
 - clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/upscli_ssl_caps.txt: upsclient, nut-client, docs: introduce UPSCLI_SSL_CAPS_CERTHOST* [#3331]
 - tests/NIT/nit.sh: consider if we can not do CERTHOST validation [#3331]
 - clients/upsclient.c: report if we can not do CERTHOST(name) validation [#3331]
 - Makefile.am: .libs-dev-PATH: fix typo
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2.86 2026-04-28 Jim Klimov <jimklimov+nut@gmail.com>

- clients/Makefile.am, common/Makefile.am, scripts/obs/debian.rules, scripts/obs/nut.spec, tools/nut-scanner/Makefile.am: **/Makefile.am, scripts/obs/**: bump libnutclient ABI level [#1290, #3331]
- clients/nutclient.cpp: drop obsolete FIXME comments [#3331]
- clients/nutclient.cpp: with openssl builds, do not require non-trivial certhost_name IF we do not have methods to check it (openssl too old) [#3331]
- tests/NIT/nit.sh: refactor by use of check_NIT_certs_NSS() [#1711]
- tests/NIT/nit.sh: introduce check_NIT_certs_NSS() [#1711]
- tests/cpptest-client.cpp: enable testing for CERTHOST with OpenSSL builds, update argument order for NSS builds [#3331, #1711]
- clients/Makefile.am, clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient_tcp.txt: clients/nutclient.{cpp,h}, docs, Makefile.am: fix SSLConfig_NSS related argument order to place certhost after certident, to match logical proximity and OpenSSL equivalents; reset ABI age (not backwards compatible to recent v2.8.5 release) [#3331]
- tests/NIT/nit.sh: refactor by use of check_NIT_certs_NSS() [#1711, #3390, #3108]
- tests/NIT/nit.sh: introduce check_NIT_certs_NSS() [#1711]
- tests/NIT/nit.sh: revise sanity-check, not all NSS versions had a *.txt file [#3331, #1711]
- tests/NIT/nit.sh: consider WITH_SSL_CLIENT_CERTIDENT=none to (re)set certificate nickname verification [#3331, #1711]
- tests/NIT/nit.sh: reword SSL support messages [#1711]
- tests/NIT/nit.sh: revise sanity-check, not all NSS versions had a *.txt file [#3331, #1711]

2.87 2026-04-27 Jim Klimov <jimklimov@gmail.com>

- clients/nutclient.cpp: fix NULL⇒nullptr [#3331]
- tests/NIT/nit.sh: isTestablePython(): fix detection of interpreter shebang via /usr/bin/env [#1711]
- tests/NIT/nit.sh: isTestablePython(): fix detection of interpreter shebang via /usr/bin/env [#1711]
- tests/NIT/nit.sh: refactor and improve portability with somehash_filter()/somehash_files() method detection [#3390]
- ci_build.sh: refactor and improve portability with somehash_filter()/somehash_files() method detection [#3108]

2.88 2026-04-27 Jim Klimov <jimklimov+nut@gmail.com>

- .circleci/config.yml: allow various cache contents for same branch/job iterations [#3390]
 - tests/NIT/nit.sh: introduce check_NIT_certs() to ensure availability of all expected certs for NSS or OpenSSL scenarios [#3390]
 - Makefile.am: .libs-dev-PATH: adjust for MacOS [#1711]
 - tests/cpptest-client.cpp: report also NUT_PORT [#1711, #3331]
 - tests/cpptest-client.cpp: fix reporting of incoming envvars [#1711, #3331]
 - clients/upsclient.c, clients/nutclient.cpp: accept systems where openssl does not define X509_V_ERR_HOSTNAME_MISMATCH [#3331]
 - clients/upsclient.c, clients/nutclient.cpp: use pragmas to hide unreachable code warnings when built against openssl without X509 checking methods we use [#3331]
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- ci_build.sh, tests/NIT/nit.sh: introduce DO_USE_NUTCI_CACHE_DEBUG to stash optional items not needed in production use of the cache, but rather to track its regressions on re-runs etc. [#3108]
- Makefile.am: use a more portable date spec in touch for start of Epoch
- configure.ac: properly use AC_CACHE_VAL for BSDKVMPROCLIBS [#3108]
- m4/nut_check_libltdl.m4: re-detect ltdl.h details if we needed to retry, to not lose track [#3108]
- tests/NIT/nit.sh: fix NUT_CAPATH for PERL SSL tests on Windows (MSYS2) yet some more [#1711]
- tests/NIT/nit.sh: fix NUT_CAPATH for PERL SSL tests on Windows (MSYS2) with forward slashes too [#1711]
- tests/NIT/nit.sh: fix NUT_CAPATH for PERL SSL tests on Windows (MSYS2) with forward slashes too [#1711]
- tests/NIT/nit.sh: fix portability for recursive "cp" [#3390]

2.89 2026-04-26 Jim Klimov <jimklimov+nut@gmail.com>

- m4/nut_check_libnetsnmp.m4: re-detect libnetsnmp details in mingw builds to not lose track [#3108]
- scripts/Windows/build-mingw-nut.sh: CI_CACHE_NUT_HASHDIR_CFG and configure_nut(): stash a copy of first-run config.log and config.h as examples of resulting configuration [#3108]
- scripts/Windows/build-mingw-nut.sh: add © into the heading
- ci_build.sh: get_CI_CACHE_NUT_HASHDIR_CFG_OPT(), configure_nut(): stash a copy of first-run config.log and config.h as examples of resulting configuration [#3108]

2.90 2026-04-25 Jim Klimov <jimklimov+nut@gmail.com>

- docs/download.txt: fix NOTE block style
 - ci_build.sh, Makefile.am: revise passing CI_CACHE_NUT_HASHDIR_CFG_OPT into distcheck runs [#3108]
 - ci_build.sh: consider_cleanup_shortcut(): do not fail if there is no out-of-tree CI_BUILDDIR/Makefile.in - normally it should not be there (but do check for one in SCRIPTDIR)
 - ci_build.sh: refactor get_CI_CACHE_NUT_HASHDIR_CFG_OPT() into a method, handle distcheck builds too [#3108]
 - NEWS.adoc: update about NIT caching of test certificates [#3390]
 - tests/NIT/README.adoc, docs/nut.dict: document caching [#3108, #3390, #1711]
 - appveyor.yml, .circleci/config.yml: generalize to DO_USE_NUTCI_CACHE, not just DO_USE_AUTOCONF_CACHE [#3108, #3390]
 - ci_build.adoc, docs/nut.dict: document caching [#3108]
 - ci_build.sh, tests/NIT/nit.sh: generalize DO_USE_NUTCI_CACHE so AUTOCONF and NIT_TESTCERT toggles may be separate; drop old logic for config.cache in build area; rename CLEAN vars to ..._BEFORE [#1711, #3390]
 - tests/NIT/nit.sh: handle DO_CLEAN_AUTOCONF_CACHE [#3390]
 - tests/NIT/nit.sh: fix preparation of PEM and JKS client key/cert from OpenSSL [#1711, #3390] Move code that landed into a wrong case.
 - tests/NIT/nit.sh: refactor preparation of PEM and JKS server key/cert from OpenSSL [#1711, #3390] Handle a situation with new openssl defaults and old(er) java keytool.
 - tests/NIT/Makefile.am, tests/NIT/nit.sh: name the default WITH_SSL_TESTS=best-effort behavior for clarity [#1711]
 - ci_build.sh, scripts/Windows/build-mingw-nut.sh: ci_build.sh: name CI_CACHE_NUT_HASHDIR with AUTOCONF prefix [#3108]
 - tests/NIT/nit.sh: prepare_NIT_certs(): rename TESTCERT_VARS.sh to .env, generate one [#3390]
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2.91 2026-04-24 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: fix NUT_CAPATH for PERL SSL tests on Windows (MSYS2) [#1711]
- tests/NIT/nit.sh: prepare_NIT_certs(): for caching, also generate NSS DB from OpenSSL PEM if we can [#3390]
- tests/NIT/nit.sh: prepare_NIT_certs(): drop obsolete comment [#3390]
- tests/NIT/nit.sh: prepare_NIT_certs(): extend validity of generated test certificates to 20 years [#3390]
- tests/NIT/nit.sh: prepare_NIT_certs(): add generation of JKS (and more PEM) from NSS, but only if we plan to cache the result [#3390]
- tests/NIT/nit.sh: prepare_NIT_certs(): wrap long command lines [#3390]
- tests/NIT/nit.sh: refactor mock cert population logic as prepare_NIT_certs(), and allow caller-provided or automatically cached mocks [#3390]
- appveyor.yml: tweak notification triggers again [#1400]
- tests/NIT/nit.sh: fix NUT_CAPATH for PERL SSL tests on Windows (MSYS2) [#1711]
- NEWS.adoc, clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, conf/upsmon.conf.sample.in, docs/man/libnutclient_tcp.txt, docs/man/upscli_add_host_cert.txt: clients/{nut,ups}client, docs: make sure CERTHOST values for FORCESSL/CERTVERIFY are used (C++), allow inheriting global setting via -1 [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: in TcpClient, track the applied SSLConfig* vs. build(/run)time ability to use NSS (and/or) OpenSSL via UPSCLI_SSL_CAPS* and consider that in getter preferences [#3331]
- clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient_commands.txt, docs/man/libnutclient_tcp.txt: clients/nutclient.{cpp,h} docs/man/libnutclient_tcp.txt: add optional CERTHOST verification to OpenSSL code path [#3331]

2.92 2026-04-23 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp: some builds manage to get into WITH_SSL_CXX without WITH_OPENSSL\NWITH_NSS defined [#3331]
- m4/nut_check_libopenssl.m4, clients/upsclient.c, clients/nutclient.cpp, server/netssl.c: openssl_cert_verify_san_name(): support OpenSSL versions before ASN1_STRING_get0_data() and potentially related helpers [#3331]
- ci_build.sh: comment about the life-cycle of config.cache file [#3108]
- ci_build.sh: align defaults for DO_USE_AUTOCONF_CACHE and DO_CLEAN_AUTOCONF_CACHE with existence of CI_CACHE_NUT_BASEDIR [#3108]
- scripts/Windows/build-mingw-nut.sh: add support for CI_TRACE envvar toggle, like in ci_build.sh
- ci_build.sh, scripts/Windows/build-mingw-nut.sh: pass CI_CACHE_NUT_HASHDIR into Windows builds if we want to and can [#3108]
- NEWS.adoc, appveyor.yml: NEWS.adoc: ability to optionally re-use a config.cache file from older runs moved to NUT v2.8.6 [#3108]
- appveyor.yml, .circleci/config.yml: cache the out-of-tree locations for config.cache [#3108]
- ci_build.sh: support reusable config.cache per "uname \+ script config opts" hash [#3108] Now even several build environments sharing a source directory with a generated configure script should be able to safely use their individual caches.
- ci_build.sh: in developer mode (no BUILD_TYPE) utilize make all-quick for less noise
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: refactor SSLConfig_CERTIDENT_NSS to not construct with a separate certstore_pass [#3331] NSS only has one pass phrase (shared for all private key database entries)

- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: refactor TcpClient and Socket to pass around SSLConfig(-derived) instances instead of private fields [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig: add getCertHostBy*() methods [#3331]
- clients/nutclient.cpp: clients/nutclient.{cpp,h}: SSLConfig_*::apply(): prepare and hide debug printouts [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig_CERTIDENT fix polymorphic-friendly reference to _certstore [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclientmem.{cpp,h}: use _empty_str in constructors, just to be sure we have a valid reference to an immortal std::string where we expect one [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig*: introduce clone() methods to own data in our sets of CERTHOST/CERTSTORE/CERTIDENT and delete it properly [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig: in the sets of CERTHOST/CERTSTORE/CERTIDENT data, retain polymorphic-friendly pointers instead of mutilated auto-casted objects [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig::addCertHost(): constrain to adding non-trivial entries only [#3331]

2.93 2026-04-22 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: move SSLConfig* method implementations from header to CPP file [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: refactor SSLConfig_NSS class to use new storage classes [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: refactor SSLConfig_OpenSSL class to use new storage classes [#3331]
- clients/nutclient.h: extend base SSLConfig class to keep track of optional CERTSTORE, CERTIDENT and/or CERTHOST info [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: introduce SSLConfig* classes to keep CERTSTORE locations and credentials, and CERTIDENT and CERTHOST info [#3331]
- clients/nutclient.h: streamline code style for SSLConfig* classes [#3331]
- clients/nutclient.cpp: Socket() constructor should not rely on openssl_cert_verify_data struct to be pre-zeroed [#3331] OBS builds, and later valgrind, confirmed an issue with random decisions and free()'s based on random data in that struct. Apparently C differs from C++ in that aspect?..

2.94 2026-04-21 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/usbhid-ups.c, NEWS.adoc: reconnect_ups(): report success more visibly [#3423] Also fix setting "quiet" driver status when we actually remain in reconnecting state (or rather will soon re-enter it because situation is not fixed)
- clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: openssl_cert_verify_san_name(): revise loop based on return from *printf() [#3331]
- clients/nutclient.cpp, server/netssl.c: fix CI warnings [#3331]
- UPGRADING.adoc, clients/nutclient.cpp, clients/nutclient.h, docs/man/Makefile.am, docs/man/libnutclient_tcp.txt, tests/NIT/nit.sh, tests/cpptest-client.cpp: clients/nutclient.{cpp,h}, tests/cpptest-client.cpp, nit.sh, docs: a CERTHOST address was missing (and subject/nickname is not the host name/url that NSS checks for) [#3331, #1711]
- clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient_tcp.txt: clients/nutclient.{cpp,h}: rename "forcessl" to match upsclient.c, and "tryssl" to not stand out [#3331]

- tests/cpptest-client.cpp: honour NUT_DEBUG_LEVEL by enabling (Socket Connect) debug toggle [#1711]
- tests/NIT/nit.sh: support passing individual NUT_DEBUG_LEVEL_{PERL,PYTHON,CPPNIT,NUTSCAN} [#1711]
- server/netssl.c, clients/upsclient.c, clients/nutclient.cpp: align messages from NSS AuthCertificate() callbacks [#3331]
- clients/nutclient.cpp: Socket::startTLS(): if NSS handshake fails, add ssl_url to the exception message [#3331]
- clients/nutclient.cpp: setSSLConfig_*(): debug-log the code path actually taken [#1711]
- clients/nutclient.cpp: Socket: document the SSL fields better [#3331]
- clients/nutclient.cpp: TcpClient::connect(): now we can config run-time for both NSS and OpenSSL, most options are shared and built-in abilities drive the final choice [#3331]
- tests/cpptest-client.cpp: setupClientSSL(): rearrange reporting of enabled SSL options [#1711]

2.95 2026-04-20 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: setenv_ssl_cppnit(): provide NUT_CERTFILE and NUT_KEYFILE values for OpenSSL tests [#1711, #3331]
 - clients/nutclient.cpp: transplant openssl_cert_verify_callback() et al from upsclient.{c,h} [#3331]
 - clients/nutclient.cpp: Socket::startTLS(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
 - clients/nutclient.cpp: fix syntax for NSS builds [#3331, #1711]
 - configure.ac: fix the checks for enabling SSL support in C+\+ builds [#1711, #3331]
 - server/netssl.c: ssl_init(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
 - server/netssl.c, server/nut_ctype.h: transplant openssl_cert_verify_callback() et al from upsclient.{c,h} [#3331]
 - clients/upsclient.c, clients/upsclient.h: clients/upsclient.{c,h}: revise code/comment markup for recent changes [#3331]
 - tests/NIT/nit.sh: ensure SAN aliases (DNS and IP) that should NOT match the current host [#1711]
 - clients/upsclient.c: upscli_init2(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
 - clients/upsclient.c: openssl_cert_verify_san_name(): use X509_check_host()/X509_check_ip_asc() where available [#3331]
 - clients/upsclient.c: openssl_cert_verify_callback(): consistently report depth along with CN (counterpart, CA...) [#3331]
 - clients/upsclient.c: openssl_cert_verify_callback(): do not call and log openssl_cert_verify_san_name() when already OK per caller [#3331]
 - clients/upsched.c: hush down byte-by-byte read tracing (away from common debug level 6)
 - clients/upsclient.c, clients/upsclient.h: clients/upsclient.c: convert openssl_cert_print_san_name() into openssl_cert_verify_san_name [#3331] Also persist openssl_cert_verify_data_t in UPSCONN_t so it is always available to callbacks when needed for this ups instance.
 - clients/upsclient.c: openssl_cert_print_san_name(): handle IPv4/IPv6 addresses in SAN [#3331]
 - clients/upsclient.c: rename example "mydata" into "openssl_cert_verify_data" (types, vars) and example methods to print/verify certs [#3331]
 - clients/upsclient.c, configure.ac: import openssl doc examples for certificate reporting/inspection into NUT styling and logging [#3331]
 - clients/upsclient.c: fix markup about switch() so it is consistent in the file [#3331]
 - clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: server/netssl.c, clients/upsclient.c, clients/nutclient.cpp: when we verify subject_CN[], consider comma as a separator too [#3331]
 - clients/nutclient.cpp: fix a missed static_cast [#3331]
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2.96 2026-04-19 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp: be sure to include openssl/x509v3.h for methods we use [#3331]

2.97 2026-04-18 Tom Eldon <tom@tomeldon.com>

- drivers/apcmicrolink.c, drivers/apcmicrolink.h: drivers/apcmicrolink.c: minor handshake fix, allow flexible frame size Change to allow flexible frame size (upto 256 bytes) by waiting for page0, as some devices with payload size != 35 were incompatible. Allow variable length serial number string during handshake (as some devices use 14 characters).

2.98 2026-04-18 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/apc_modbus.c: addvar() the modbus_retries option [#3414]

2.99 2026-04-16 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: setenv_ssl_common(): revise NUT_CAPATH conditions [#1711]
- tests/NIT/nit.sh: fix check of python FORCESSL is testable [#1711, #1349]

2.100 2026-04-15 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: fix sleep() [#1349, #1711]
 - clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/libnutclient_tcp.txt, docs/man/upscli_ssl_caps.txt, server/netssl.c, tests/NIT/nit.sh: Further differentiate reported CERTIDENT_NAME/_PASS capability [#1711]
 - tests/NIT/nit.sh: refactor with setenv_ssl_common(), and in setenv_ssl_python() verify that we can run SSL tests (else do not force them) [#1349, #1711]
 - scripts/python/module/PyNUT.py.in: detect inability to use SSL module (older interpreter etc.) [#1349, #1711]
 - tests/NIT/nit.sh: align generation of CERTIDENT into upsd.conf/upsmon.conf with ability announced by the binaries [#1711]
 - scripts/perl/test_nutclient.pl: ANSI colors are a nice to have, do not crash if absent [#1711]
 - scripts/perl/UPS/Nut.pm: StartTLS(): use a better portable "or" for "undef" printing [#1711]
 - scripts/perl/test_nutclient.pl: warnings are a nice to have, do not crash if absent [#1711]
 - NEWS.adoc, clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/libnutclient_tcp.txt, docs/man/upscli_ssl_caps.txt: clients/upsclient.c, clients/nutclient.cpp, headers, docs: propagate UPSCLI_SSL_CAPS_CERTIDENT as a flag [#3331]
 - server/netssl.c, clients/upsclient.c: *ssl_caps_descr(): mention OpenSSL CERTIDENT (in)ability [#3331]
 - scripts/perl/UPS/Nut.pm: warnings are a nice to have, do not crash if absent [#1711]
 - clients/upsclient.c, clients/nutclient.cpp, server/netssl.c, m4/nut_check_libopenssl.m4: for OpenSSL password callback, check actual presence of methods, not version numbers [#3331]
 - clients/upsclient.c: nss_password_callback(): reconcile markup with server/netssl.c [#3331]
 - server/netssl.c: for now openssl_password_callback() should only be declared/implemented when OPENSSL_VERSION_NUMBER >= 1.1.0 [#3331]
 - tests/NIT/nit.sh: account SKIPPED test cases (when isTestable* says we can not) [#1711]
 - tests/NIT/nit.sh: be portable, use TABCHAR instead of "\t" [#1711]
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- docs/nut.dict: update for older systems
- configure.ac: work around ancient autotools without PACKAGE_URL
- Makefile.am: hijack PARMAKES_OPT to also set V=0 by default

2.101 2026-04-14 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: drivers/apcmicrolink.c: try to use naming from apc_modbus for some commands and values
- drivers/apcmicrolink.c: fix microlink_page0_state_t init and reset

2.102 2026-04-14 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: for now openssl_password_callback() should only be declared/implemented when OPENSSL_VERSION_NUMBER >= 1.1.0 [#3331]
- conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: clarify that CERTIDENT support requires OpenSSL 1.1.0\+ to fully work at the moment [#3331]
- drivers/nutdrv_qx.c, NEWS.adoc: qx_is_usb_device_supported(): only claim SUPPORTED if subdriver_command was assigned [#3410]

2.103 2026-04-14 Axel Gembe <axel@gembe.net>

- NEWS.adoc, drivers/apc_modbus.c: apc_modbus: increase default TCP response timeout to 2000 ms The default 500ms timeout value is marginal for Modbus TCP, even on a fast local network. In tests with an APC UPS, we see timeouts because of this: `` Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_read_registers: Read of 0:27 failed: Connection timed out (192.168.0.102:502) Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_read_registers: Read of 0:27 failed: Invalid data (192.168.0.102:502) Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_close: Closing connection Apr 14 19:10:47 ares apc_modbus[2002143]: Opened modbus successfully `` Packet capture: `` No. Time Source Destination Protocol Length Info 3636 2026-04-14 19:10:43.842072 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3637 2026-04-14 19:10:44.018369 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11905 Win=3660 Len=0 3638 2026-04-14 19:10:44.377723 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12623; Unit: 1, Func: 3: Read Holding Registers 3639 2026-04-14 19:10:44.578103 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11917 Win=3648 Len=0 3640 2026-04-14 19:10:44.579449 192.168.0.102 192.168.0.40 Modbus/TCP 117 Response: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3641 2026-04-14 19:10:44.579482 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [ACK] Seq=11917 Ack=70504 Win=63 Len=0 3642 2026-04-14 19:10:44.579604 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [FIN, ACK] Seq=11917 Ack=70504 Win=63 Len=0 `` 3636 is the first read, 3638 is the retry after 500ms without response, both are ACKed. 3640 is the actual response to transaction 12622, which arrives 700ms after the initial packet. At this point we expect a response to transaction 12623, which is why we fail with "invalid data". To fix this we increase the default TCP response timeout to 2000 ms, which seems to be more reasonable for a device that already can take 700 ms to respond under ideal conditions. I chose a bit higher value for users who might connect the device over a higher latency network.
- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode RunTimeCalibrationStatus_BF RunTimeCalibrationStatus_BF is explained in MPAO-98KJ7F_R1_EN Appendix B. This adds defines for the bitfield and decodes it into experimental.ups.calibration.result using a string join converter.
- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode PowerSystemError_BF as alarms PowerSystemError_BF is explained in MPAO-98KJ7F_R1_EN Appendix B. This adds defines for the bitfield and decodes it as alarms in the driver.
- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode BatterySystemError_BF as alarms BatterySystemError_BF is explained in MPAO-98KJ7F_R1_EN Appendix B. We were already using the bitfield for the RB status, this adds defines for the bitfield and decodes it as alarms in the driver.

- NEWS.adoc, docs/man/apc_modbus.txt, drivers/apc_modbus.c: apc_modbus: simplify error handling with read retries Replace the complex `_apc_modbus_handle_error` function with a simpler retry mechanism built into `_apc_modbus_read_register`. The new approach: - Retries register reads on ETIMEDOUT errors up to `modbus_retries` times (configurable, default 3) - On non-timeout errors or after retry exhaustion, closes the connection for reconnection on the next update cycle - Removes the platform-specific (WIN32/POSIX) timeout detection and the flush-based recovery that didn't work anyway (flush is already done in `_apc_modbus_reopen` upon reconnection) Also adds a `modbus_retries` driver option to configure the number of read retry attempts, and improves logging for connection open/close events. This change was inspired by a patch by @marcan to do the same and from testing the behaviour of `apcupsd`, noticing that on my USB unit it times out on read and only succeeds on the first retry.
- NEWS.adoc, drivers/apc_modbus.c: apc_modbus: always zero terminate string join `_apc_modbus_string_join` did not zero terminate the result in every case, like when all values are NULL.
- drivers/apc_modbus.c: apc_modbus: bump driver version to 0.20

2.104 2026-04-13 Lukas Schmid <lukas.schmid@netcube.li>

- docs/nut.dict: Add more words to dict
- drivers/apcmicrolink.c: fix most static compile issues
- docs/nut.dict: add apcmicrolink and RMI

2.105 2026-04-13 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: introduce apcmicrolink driver [#3406]
- drivers/Makefile.am: EXTRA_DIST apcmicrolink-maps.h apcmicrolink.h [#3406]
- docs/man/Makefile.am: provide apsmicrolink.html also [#3406]
- tests/NIT/nit.sh: generate upsmon.pem [#3331, #1711]
- NEWS.adoc, clients/nutclient.cpp, clients/upsclient.c, conf/upsd.conf.sample, conf/upsmon.conf.sample.in, docs/man/upsd.conf.txt, docs/man/upsmon.conf.txt, server/netssl.c: upsd, libupsclient, libnutclient, upsmon: in OpenSSL builds, support CERTPATH pointing to a big PEM file as well as a directory [#3331]
- clients/nutclient.cpp, clients/nutclient.h, tests/cppptest-client.cpp: clients/nutclient.{h,cpp}, tests/cppptest-client.cpp: extend `setSSLConfig_OpenSSL()` etc. to support CERTIDENT name (validation of client's own cert) [#3331]
- clients/upsclient.c: fix CERTIDENT (OpenSSL) to check for CN=... part too [#3331]
- server/netssl.c: fix CERTIDENT (OpenSSL) to check for CN=... part too [#3331]
- tests/NIT/nit.sh: generate upsmon.pem [#3331, #1711]
- tests/NIT/nit.sh: `setenv_ssl_cppnit()`: fix check for TESTCERT_PATH_ROOTCA [#1711]
- tests/NIT/nit.sh: unlock OpenSSL features that are now on par with NSS [#3331, #1711]
- scripts/python/module/PyNUT.py.in, scripts/perl/UPS/Nut.pm: handle CERTIDENT to validate loaded client certificate name [#3331] Make sure the file we got is the file we wanted. Belts and suspenders rule!
- scripts/python/module/PyNUT.py.in, scripts/perl/UPS/Nut.pm: handle CERTHOST (dict or list of dict/tuple) to validate connections better [#3331]
- scripts/perl/UPS/Nut.pm: `StartTLS()`: alias variables named like in other NUT code into `SSL_*` if not provided directly [#1711]
- clients/nutclient.cpp, clients/upsclient.c, conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/netssl.c: clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: reconcile variants of `openssl_password_callback()` [#3331] Notably, in C code (upsclient, netssl) constrain with `OpenSSL >= 1.1` support, and do use `SSL_CTX_set_default_passwd_cb_userdata()` to pass the configured (or generally otherwise provided) `certpasswd` string.

- NEWS.adoc, clients/upsmon.c, conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: clients/upsmon.c, NEWS.adoc, conf/, **docs/man/**: introduce support of CERTFILE to upsmon [#3331]
- clients/upsclient.c, clients/upsclient.h, docs/man/Makefile.am, docs/man/upscli_init.txt: clients/upsclient.{c,h}, docs/man/upscli_init.txt: implement client CERTIDENT correctly (via CERTFILE); introduce upscli_init2() [#3331]
- NEWS.adoc, conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/conf.c, server/netssl.c, server/netssl.h: server/*, NEWS.adoc, conf/upsd.conf.sample, docs/man/upsd.conf.txt: implement server-side CERTPATH for OpenSSL [#3331] Optionally use for validation of clients with OpenSSL collection of CA PEM files.
- conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/conf.c, server/netssl.c: server/*, conf/upsd.conf.sample, docs/man/upsd.conf.txt: implement server-side CERTIDENT for OpenSSL [#3331]
- NEWS.adoc, conf/upsd.conf.sample, conf/upsmon.conf.sample.in, docs/man/upsd.conf.txt, docs/man/upsmon.conf.txt: NEWS.adoc, docs/man/, **conf/**: update docs about newly supported features with OpenSSL [#3331]
- server/conf.c: reshuffle options processing and update comments now that CERTREQUEST is for both backends and more may follow [#3331]
- clients/upsclient.c: upscli_sslinit(): consider CERTHOST→certverify setting to NOT abort the connection if verification was not required (0) [#3331]
- clients/upsclient.c: upscli_sslinit(): consider OPENSSL_VERSION_NUMBER constraint in logging, possibly rejecting the StartTLS attempt with OpenSSL too old (can not fulfil required security, avoid sad mishaps) [#3331]
- conf/upsd.conf.sample, docs/man/upsd.conf.txt: heads-up about MAXCONN and ulimit [#3365]
- NEWS.adoc: remove duplicated paragraph (silent merge conflict)
- clients/upsmon.c, clients/upsrw.c, common/common.c, drivers/adelsystem_cbi.c, drivers/generic_modbus.c, drivers/main.c, drivers/powerman-pdu.c, drivers/snmp-ups.c, drivers/triplite_usb.c, tools/nut-scanner/nutscan-serial.c: **.c: fix use of non-const char** with strstr() and strchr() return values [#823]
- docs/man/Makefile.am: fix possible non-delivery of NUTSCAN_UPSLOG_SET_DEBUG_LEVEL_DEPS pages [#3378, #3379]
- docs/man/upscli_init.txt: update wording and markup for pre-existing features [#3331]
- NEWS.adoc: announce the adjusted markup format

2.106 2026-04-12 Jim Klimov <jimklimov@gmail.com>

- scripts/perl/UPS/Nut.pm: warn if CERTVERIFY and custom CA options are enabled on Darwin platform [#3404]
 - scripts/perl/UPS/Nut.pm: align closer with can_ssl [#1711]
 - scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl, tests/NIT/nit.sh: scripts/perl/*, tests/NIT/nit.sh: streamline passing NUT_DEBUG (or defaulting to no debug) [#1711]
 - tests/NIT/nit.sh: neuter CERTVERIFY on darwin for now [#3404]
 - tests/NIT/nit.sh: centralize definition of PERL_OPTS [#1711]
 - scripts/perl/UPS/Nut.pm: allow SSL cert validation for IP addresses as host names [#1711]
 - scripts/perl/UPS/Nut.pm: pass only defined args to IO::Socket::SSL→start_SSL() [#1711]
 - scripts/perl/UPS/Nut.pm: support debugging messages of IO::Socket::SSL [#1711]
 - tests/NIT/nit.sh: report the non-default NIT_CASE being handled [#1711]
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2.107 2026-04-12 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: upscli_sslinit(): OPENSSL_VERSION_NUMBER constraint applies to whole block for X509_VERIFY_PARAM [#3331]
- server/netssl.c: extend OpenSSL builds feature set to support CERTREQUEST validation of clients [#3331]
- clients/upsclient.c: extend OpenSSL builds feature set to support pinning server certificates [#3331]
- clients/upsclient.c: extend OpenSSL builds feature set to support a client certificate\+key, optionally with password [#3331]
- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/UPS/Nut.pm: now we can pass SSL debug level as a number [#1711]
- scripts/perl/UPS/Nut.pm: fix to loading IO::Socket::SSL via eval with quotes not braces [#1711] It is then evaluated at run-time as code reaches that line (or not - conditionals), not at interpretation time where it fails too early. Also use "1;" in the end as a tidy practice.
- clients/nutclient.cpp: work around lack of threads in some mingw versions [#3402]

2.108 2026-04-11 Lukas Schmid <lukas.schmid@netcube.li>

- data/driver.list.in, docs/man/Makefile.am, docs/man/apcmicrolink.txt, drivers/Makefile.am, drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c, drivers/apcmicrolink.h: drivers: add experimental APC Microlink serial driver Implement a new apcmicrolink driver for APC Smart-UPS units that expose the Microlink serial protocol. The driver establishes a Microlink session over the serial link, reads the device descriptor blob, and uses that runtime descriptor map to publish standard NUT variables, writable settings, status bits, alarms, and supported instant commands. Add labeled Microlink command masks for battery tests, calibration, UPS control, and outlet-group power control; wire APC-style load and shutdown commands through the static command map; publish input.transfer.reason from UPS status change cause; and stop folding simple-signaling status bits into ups.status. Also hook upsdrrv_shutdown() up to the Microlink shutdown request, document the driver, and clean up initial upstream review issues such as missing file headers, whitespace, and format-string warnings.

2.109 2026-04-11 Jim Klimov <jimklimov+nut@gmail.com>

- m4/ax_c_pragmas.m4, clients/nutclient.cpp: introduce HAVE_PRAGMA_GCC_DIAGNOSTIC_IGNORED_DEPRECATED_DECLARATIONS to implement methods for deprecated "master" operation [#840, #3402]

2.110 2026-04-10 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: announce updates to client binding libraries [#3402]
 - scripts/perl/test_nutclient.pl: avoid "Can't call method "Error" on an undefined value" messages in failed tests; clearly say that \$nut is no more [#1711]
 - scripts/perl/UPS/Nut.pm: when IO::Socket::SSL is not available, report the error with "FEATURE-NOT-SUPPORTED" in text [#1711]
 - tests/NIT/nit.sh: setenv_ssl_perl(): check if IO::Socket::SSL is available, neuter NUT_FORCESSL based on that [#1711]
 - scripts/perl/test_nutclient.pl: support USESSL=undef if no value provided in env [#1711]
 - scripts/perl/UPS/Nut.pm: support USESSL=undef, check that we can_ssl and follow up with that [#1711]
 - scripts/perl/UPS/Nut.pm: _initialize(): include previously reported error in SSL setup fail messages [#1711]
 - scripts/perl/UPS/Nut.pm: apply indentation consistently (3 styles were intermixed)
 - scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: support NUT_SSL envvar toggle to disable attempts at SSL altogether [#1711]
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- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: SSL_ca_path should be defined in args by caller, not by a hack in production code [#1711]
- scripts/perl/UPS/Nut.pm: STARTTLS: debug-print %arg list that is passed to the SSL library [#1711]
- scripts/perl/UPS/Nut.pm: use CAPATH to provide SSL_ca_path, not SSL_ca_file [#1348, #1711]
- scripts/python/module/PyNUT.py.in: retry SSL handshake if it timed out, a few times [#3401]
- server/netssl.c: net_starttls(): report which SSL backend is in place (if any) [#3331]
- scripts/python/module/PyNUT.py.in: isValidProtocolVersion(): stringify and strip the "result" before matching in regex [#1349]
- clients/Makefile.am: bump libupsclient ABI version
- tests/NIT/nit.sh: add a note about jNut using this script [#1711, #1350]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, tests/cpptest-client.cpp: clients/nut-client{,mem}.{h,cpp}, tests/cpptest-client.cpp: remember if we enabled TRACKING so it is really done once, confirm with isTrackingModeEnabled() [#656]
- scripts/perl/test_nutclient.pl: fix DeviceLogin⇒ListClient for "admin" without a role [#1711]
- scripts/perl/UPS/Nut.pm: _send(): reset the error buffer before networking away [#1711, #1348]
- scripts/perl/UPS/Nut.pm: we can only authenticate once per connection, skip other attempts [#1711, #1348]
- docs/download.txt: publish jNut-1.1 [#1350]
- NEWS.adoc: announce updates to client binding libraries [#3402]
- appveyor.yml: post only one (custom) notification per build [#1400]
- docs/man/Makefile.am: hush "cat" errors in LIBNUTCLIENT_TCP_DEPS_SUB recipe

2.111 2026-04-09 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/perl/test_nutclient.pl: fix reporting of @failed methods [#1711]
 - scripts/perl/UPS/Nut.pm: GetTrackingResult(): fix parsing results of GET TRACKING [#1348]
 - tests/NIT/nit.sh: when we ask for specific NIT_CASE values (cppnit, python, perl, nutscanner) the lack of prerequisites should be a FAILURE [#1711]
 - tests/NIT/nit.sh: introduce tests for PERL [#1348, #1711]
 - message, so we can parse it [#1348, #1711]
 - scripts/perl/test_nutclient.pl: fix access to possibly undefined envvars [#1348, #1711]
 - scripts/perl/UPS/Nut.pm: fix error messages when Authenticate() and Login() fail [#1348, #1711]
 - scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/test_nutclient.pl: fix test case setup, update comments [#1711, #1348]
 - scripts/perl/UPS/Nut.pm: fix string work with TRACKING logic
 - scripts/perl/UPS/Nut.pm: change commands that address an UPS to optionally accept the UPS name as argument, only using
 - scripts/python/module/test_nutclient.py.in: fix typo in debug traces
 - assignments into debug() calls [#1348, #1711]
 - COPYING, scripts/HP-UX/nut.psf.in, scripts/Makefile.am, scripts/obs/debian.libups-nut-perl.install, scripts/perl/{ ⇒ UPS}/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/test_nutclient.pl: introduce tests, move Nut.pm to UPS subdir [#1348, #1711]
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- scripts/perl/Nut.pm: call StartTLS() from _initialize(), and use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387, #1348]
- clients/nutclient.cpp, clients/upsclient.c, scripts/python/module/PyNUT.py.in: Client libraries (C, C++, Python): use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, clients/upsclient.c, clients/upsclient.h, scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: Client libraries (C, C++, Python, PERL): introduce an isValidProtocolVersion() method to actively ping a connection [#3387]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
- scripts/python/module/PyNUT.py.in: refactor FSD() with an explicit becomePrimary() method
- scripts/perl/Nut.pm: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
- scripts/perl/Nut.pm: add warnings toggle
- scripts/perl/Nut.pm: fix typos in comments
- scripts/perl/Nut.pm: rename primary()⇒becomePrimary, add becomeSecondary() for completeness [#1348]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: rename primary()⇒becomePrimary, add becomeSecondary() for completeness [#656]
- tests/cpptest-client.cpp: add test with SET VAR TRACKING [#1711, #656]
- scripts/python/module/test_nutclient.py.in: add test with SET VAR TRACKING [#1711, #1349]
- scripts/perl/Nut.pm: fix typos in comments
- docs/download.txt: refer to new release of jNut-1.0 (and the following snapshot evolution)
- tests/NIT/nit.sh: do not unset DEBUG_SLEEP when we generate configs [#1711] We might actually want more of those, with test case passwords revealed, etc.

2.112 2026-04-08 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: TrackingID class: include timestamp tracking and age reporting [#1349]
 - scripts/perl/Nut.pm: TrackingID class: include timestamp tracking and age reporting [#1348]
 - clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}, tests/*.cpp: implement waiting for TRACKING result if requested [#656]
 - clients/nutclient.h, clients/Makefile.am: Implement TrackingID as a separate class, including timestamp tracking and age reporting [#656]
 - scripts/python/module/PyNUT.py.in: Implement TrackingID as a separate class [#1349]
 - scripts/python/module/PyNUT.py.in: fix syntax [#1349]
 - scripts/perl/Nut.pm: Implement TrackingID as a separate class [#1348]
 - scripts/perl/Nut.pm: fix syntax [#1348]
 - scripts/python/module/PyNUT.py.in: revise AI creations, complete the TRACKING\+waiting dialog support [#1349]
 - scripts/perl/Nut.pm: revise AI creations, complete the TRACKING\+waiting dialog support [#1348]
 - scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: AI PoC: implement the remainder of NUT Networked protocol (as of NUT v2.8.5) for Python and Perl bindings
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- scripts/perl/Nut.pm: clean up trailing white-space
- tests/NIT/nit.sh: simplify with NIT_CASE=generatecfg_sandbox support [#1711]
- tests/NIT/nit.sh: allow injecting WITH_SSL_CLIENT, WITH_SSL_SERVER, WITH_SSL_SERVER_CLIVAL (e.g. to test against older packaged builds) [#1711]
- .github/workflows/05-codeql.yml: drop duplicate init stage
- configure.ac: hush clang-21 "-Wc11-extensions" warnings about system headers
- docs/download.txt: fix NOTE markup for N4W 7Zip size

2.113 2026-04-07 Jim Klimov <jimklimov+nut@gmail.com>

- docs/maintainer-guide.txt: revise instructions for post-release steps (website, etc.)
- docs/download.txt: newly refer to nut-X.Y.Z-docs.tar.gz archives
- docs/docinfo.xml.in: fix release date of v2.8.5 and move its tag to be above v2.8.4
- NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: Revert "NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: finalize text before NUT v2.8.5 release" This reverts commit 784219a3dc935e53a1273063c591bc6dd949590b: finally released NUT v2.8.5, preparing for v2.8.6 cycle now.

2.114 2026-04-06 Jim Klimov <jimklimov+nut@gmail.com>

- common/wincompat.c: syslog(), send_to_named_pipe(): debug-log when we failed to open existing NAMED_PIPE and so move on [#3302]
- common/wincompat.c: log errors after methods faults via upslog_with_errno() not plain upslogx() [#3302]
- common/wincompat.c: log progress through pipe*() methods [#3302, #3368]
- common/wincompat.c: fix send_to_named_pipe() to measure strlen(data) once [#3302]
- common/wincompat.c: fix getpass() to measure strlen(wincompat_password) once [#3302]
- common/wincompat.c: fix filter_path() to measure strlen(source) once [#3302]
- common/wincompat.c: tcflush(): fix bitwise OR to boolean OR

2.115 2026-03-31 Jim Klimov <jimklimov+nut@gmail.com>

- server/upsd.c: sendback(): adjust debug traces for write() vs ssl_write() [#3379, #3331]
- server/netssl.c: net_starttls(): send_err_extra(NUT_ERR_ACCESS_DENIED) for faults in NSS setup [#3379, #3331]
- server/upsd.c, server/upsd.h: server/upsd.{c,h}: introduce send_err_extra() [#3331]
- drivers/dstate.c, NEWS.adoc: revise closing of connections and returning error codes; document the methods [#3302]

2.116 2026-03-30 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/dstate.c: debug-log the progress around starting and maybe failing CreateNamedPipe() [#3302]
 - drivers/dstate.c: sock_disconnect(): resync code with wincompat::pipe_disconnect() implementation [#3302]
 - drivers/dstate.c: sock_disconnect(): try to flush the buffer [#3368]
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2.117 2026-03-25 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/dstate.c: err on the safe side and set conn=NULL after sock_disconnect() [#3368]

2.118 2026-03-24 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/upsdrvctl.c: refactor with forkexec_parent_analyze() and a re-check before retry - also for WIN32 [#3302] Also revised WaitForSingleObject() result checking - there has to be a chance to succeed ;)
- drivers/upsdrvctl.c: refactor with forkexec_parent_analyze() and a re-check before retry - POSIX part [#3302]

2.119 2025-10-02 Jim Klimov <jimklimov+nut@gmail.com>

- configure.ac: WIN32 GetAdaptersInfo(): be careful about \$NETLIBS_GETADDRS usage
 - ci_build.sh: retain config.cache if DO_CLEAN_AUTOCONF_CACHE=no also in dev-test builds [#3108]
 - ci_build.sh: configure_nut(): log if config.cache is available and if we want/do use it [#3108]
 - ci_build.sh: do not default to DO_USE_AUTOCONF_CACHE=yes/DO_CLEAN_AUTOCONF_CACHE=no [#3108] Callers like specific CI recipes should be free to enable certain behaviors though.
 - .circleci/config.yml: default to DO_CLEAN_AUTOCONF_CACHE:no DO_USE_AUTOCONF_CACHE:yes and carry over the config.cache between builds [#3108]
 - appveyor.yml: default to DO_CLEAN_AUTOCONF_CACHE:no DO_USE_AUTOCONF_CACHE:yes and carry over the config.cache between builds [#3108]
 - ci_build.sh, scripts/Windows/build-mingw-nut.sh, NEWS.adoc: introduce options to USE_AUTOCONF_CACHE and/or CLEAN_AUTOCONF_CACHE [#3108]
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